Rank-seven Grothendieck rings: certified exclusions and models through multiplicity four

Abstract

We study complex categorification of the complete rank-seven fusion-ring census through multiplicity four. Exact necessary conditions exclude 4164 rings, and complete tensor comparisons identify unitary models for 27 more; 10 remain unresolved. The theoretical arguments combine prime equivariantization, pointed-module and orbit obstructions, with complete signed-sphericalization certificates and a relative-trace obstruction that needs no signed enumeration. A cyclic-stabilizer divisibility argument excludes five further rings, while a three-parameter argument forces sphericality for two rings with a dimension-block obstruction. Independent signed spectral certificates verify two of the cyclic-stabilizer exclusions. A full induced-unit compression bound yields a further general exclusion by a parity contradiction. Anti-invariant module spectra and Morita-dual Gram identities supply eighteen further general exclusions, twelve by elementary theoretical arguments and six by exact finite certificates. Two Morita transformations and a formal-codegree field obstruction give three more general exclusions. Multiplicities one and two are complete, with twenty-three realized rings. Every counted general exclusion applies without a pivotality assumption, and every positive row has an actual categorical construction.

1 Scope and statements

Let R have distinguished basis $(b_0 = 1, b_1, \dots, b_6)$ and multiplication $b_i b_j = \sum_k N_{ij}^k b_k$. A *Grothendieck ring* here means the based ring of a complex fusion category; no pivotal, spherical, braided, or unitary hypothesis is placed on a hypothetical categorification. Its multiplicity is $\max_{i,j,k} N_{ij}^k$. The upstream *Fusion Ring Census* [14] provides the complete rank-seven fusion-ring list through multiplicity four. We pin its source bytes, source commit, row ordering, and their digest in the companion package. This is a categorification study of that census; the list of undecided tensors is not counted as a list of Grothendieck rings. Multiplicities at least five lie outside the pinned source and the claims of this manuscript.

Theorem 1 (Certified rank-seven status through multiplicity four)**.** *The numbers of excluded, explicitly unitarily realized, and undecided rank-seven fusion rings are given by Table 1. The negative conclusions hold for* every *complex fusion category. The twenty-seven positive rows have the models in Table 3, matched against their complete fusion tensors. No categorifiability conclusion is drawn for the undecided rows.*

Table 1: Certified status of complete rank-seven fusion-ring census through exact multiplicity four. Undecided entries are not assumed categorifiable.

Exact multiplicity	Fusion rings	Excluded	Realized	Undecided
1	43	30	13	0
2	319	309	10	0
3	1059	1054	2	3
4	2780	2771	2	7

Table 2: Sequential exclusion counts, charged only to the first successful criterion. The later rows record additional theoretical and exact finite obstructions.

Criterion	$m = 1$	$m = 2$	$m = 3$	$m = 4$
Near-group theorem	2	1	1	1
Global dimension 15	1	0	0	0
Zero spectrum	8	30	51	63
Closed rank-four/five subring	0	25	125	235
Formal-codegree d-number	15	179	788	2314
Subcategory dimension quotient	0	5	9	8
Oriented dimension order	4	51	61	130
Localized triangular-prism equations	0	1	1	0
Solvability and support inequality	0	0	2	0
Half-partition Tannakian criterion	0	0	1	0
Odd pointed group in dimension 120	0	0	0	1
Double equivariantization in dimension 52	0	1	0	0
Order-sixteen orbit obstruction	0	2	0	0
Dimension-96 rank-one modules	0	2	0	0
Dimension-$8p$ equivariantization	0	2	0	1
Dimension-96 order-sixteen orbits	0	2	0	0
Dimension-112 Singer normalizer	0	0	2	0
Dimension-54 Brauer–Picard	0	0	2	0
Additional complete signed quotients	0	0	2	2
Dimension-310 group theory	0	0	0	1
Relative-trace lattice	0	0	0	2
Dimension-150 central boson	0	1	0	0
Cyclic-stabilizer divisibility	0	0	0	5
Signed three-parameter sphericality	0	0	2	0
Full induced-unit compression and parity	0	0	1	0
Anti-invariant module spectral bounds	0	2	3	2
Morita-dual Gram identities	0	4	2	1
Positive-semidefinite Gram refinement	0	0	1	3
C_3–A_4 Morita obstruction	0	1	0	2

Table 3: Unitary model constructions identified by exact full-tensor based-ring isomorphism. Each row denotes a ring, not a count of categories.

Census ID	Explicit model
R7_1_1	$\mathrm{Rep}(D_{16})$
R7_1_6	$\mathrm{SU}(2)_6$
R7_1_9	$\mathrm{PSU}(2)_{12}$
R7_1_12	$\mathrm{SO}(7)_2$
R7_1_16	$\mathrm{Rep}(D_{22})$
R7_1_17	$\mathrm{PSU}(2)_{13}$
R7_1_19	AnyonWiki $\mathrm{FC}^{7,1,2}_{3,1,0,1}$
R7_1_24	$\mathrm{Zest}(\mathrm{SU}(2)_6)$
R7_1_25	$\mathrm{Rep}(SD_{16})$
R7_1_28	AnyonWiki $\mathrm{FC}^{7,1,2}_{12,1,0,1}$
R7_1_36	AnyonWiki $\mathrm{FC}^{7,1,4}_{3,1,0,1}$
R7_1_39	$\mathrm{TY}(C_6)$
R7_1_43	$\mathrm{Vec}(C_7)$
R7_2_1	TY parity equivariantization (§8.1)
R7_2_24	$\mathrm{Rep}(S_5)$
R7_2_195	TY twisted-C_4 equivariantization (§8.1)
R7_2_218	$\mathrm{Rep}(C_{13} \rtimes C_4)$
R7_2_233	$\mathrm{Rep}(H_{24})$
R7_2_234	$\mathcal{C}(A_5, 1, A_4, 1)$
R7_2_240	D_{12} even category (§8.4)
R7_2_297	$\mathrm{Rep}(SL_2(\mathbb{F}_3))$
R7_2_313	$\mathrm{Adj}(\mathrm{SU}(3)_5)$
R7_2_319	$\mathrm{Rep}(C_{13} \rtimes C_3)$
R7_3_602	$\mathrm{Rep}(A_6)$
R7_3_1059	$\mathrm{Rep}(C_{11} \rtimes C_5)$
R7_4_2	$\mathrm{Vec}(Q_8)^{S_3}$
R7_4_2019	Twisted quaternion model (§8.3)

Multiplicity one is complete: its former residuals $R7_1_19$, $R7_1_28$, and $R7_1_36$ have the AnyonWiki models described below. The first undecided multiplicity is two. The exhaustive source-position convention is $R7_m_j$, where j is the position *among rows of exact multiplicity* m in the pinned census file. This convention makes the positive-model permutations and negative witnesses unambiguous.

2 Necessary conditions and their order

The checks in Table 2 run from top to bottom on surviving tensors, and a tensor is charged to its first successful condition. Near-group and small-dimension checks read a few coefficients; the zero-spectrum and closed-subring passes are finite tensor scans; exact Casimir factoring and subcategory-dimension arithmetic follow. The complete oriented search ends this initial sequence; the further theoretical exclusions are appended as a dated update. This order records precisely when a criterion first contributes rather than hiding it in an unordered union of tests. Multiplicity two is still undecided, so these stages do *not* establish a multiplicity threshold beyond which no new criteria would be needed. The oriented criterion first contributes at multiplicity one, so it cannot be deferred to a higher multiplicity without losing a proof.

If G is the group of invertible basis elements and the only other object X satisfies $X^2 = \sum_{g\in G} g + \ell X$, then for $|G| = 6$: at $\ell = 0$ the Tambara–Yamagami construction forces G abelian [29]; at $0 < \ell < 5$ in the commutative case, Schopieray's near-group obstruction rules out categorification [28, Proposition 3.2]. These statements apply to a categorification without a pivotal assumption. The certificate records the six group elements, the remaining object and ℓ, and rechecks every fusion-rule hypothesis.

The global-dimension-fifteen branch has six dimension-one simples and one dimension-three simple: direct substitution into every product establishes that $(1, \dots, 1, 3)$ is its strictly positive dimension character. Hence its Frobenius–Perron dimension is $6 + 3^2 = 15$. The pq classification [19, Theorems 3.1 and 5.1] makes a complex fusion category of dimension $3 \cdot 5$ pointed in this case, contradicting the dimension-three simple.

The zero-spectrum obstruction is the exact necessary criterion of Liu–Palcoux–Ren [23]; each certificate stores its nine basis indices and the verifier rechecks all required zero/nonzero fusion coefficients. A proper fusion-closed subset of basis elements would generate a fusion subcategory. If its induced rank-four or rank-five ring is excluded by the complete lower-rank classifications [15, 16], the ambient ring is excluded. The imported model registries contain all fifteen rank-four and all seventeen rank-five realized based-ring types, checked up to multiplicities 128 and 20, respectively; all encountered subrings fall within these ranges. The complete rank-six multiplicity-two classification [17] supplies context and methodology, without being used as an unstated rank-seven classification theorem.

Proposition 2 (The formal-codegree test at rank seven)**.** *Suppose that R is a rank-seven fusion ring, possibly noncommutative. Let $C = \sum_{i=0}^{6} N_i N_i^{\mathsf{T}}$ be its regular Casimir matrix. If an irreducible factor of the characteristic polynomial of C is not the minimal polynomial of a d-number, then R cannot be the Grothendieck ring of a complex fusion category.*

Proof. In the commutative case, the eigenvalues of C are the formal codegrees of one-dimensional characters and must be d-numbers by Ostrik [26]. In the noncommutative case, semisimplicity, $\dim_{\mathbb{C}}(R \otimes \mathbb{C}) = 7$, and the positive one-dimensional character imply the unique possible decomposition $R \otimes \mathbb{C} \simeq \mathbb{C}^3 \oplus M_2(\mathbb{C})$. There is only one two-dimensional irreducible module; it is fixed by every field automorphism, so its formal codegree is a rational algebraic integer and hence an integer. The Casimir acts on this matrix block by twice that integer. Every remaining nonrational Casimir eigenvalue comes from a one-dimensional character and is a formal codegree, so Ostrik's condition applies. The implementation factors the exact integral characteristic polynomial and tests the coefficient-divisibility characterization of d-numbers. The replay checker recomputes and factors that polynomial. □

If S is a proper commutative fusion subring of a hypothetical categorification, the quotient $\mathrm{FPdim}(R)/\mathrm{FPdim}(S)$ must be an algebraic integer [21]. The subcategory-divisibility stage computes this quotient in the exact Frobenius–Perron number field, records its characteristic polynomial, and certifies a nonintegral coefficient. Its certificate is rederived from the complete tensor.

3 Complete oriented tests at rank seven

The oriented-dimension-order row of Table 2 uses the general, *oriented* signed-sphericalization construction established in [17]. For an arbitrary hypothetical category $\mathcal{C}$, its canonical sphericalization has two lifts of each simple object, interchanged by a sign object ε. Taking the quotient by $(\varepsilon + 1)$ produces integer coefficients T_{ij}^k with $|T_{ij}^k| \le N_{ij}^k$ and $T_{ij}^k \equiv N_{ij}^k \pmod 2$. Unit, duality,

cyclic Frobenius reciprocity, and ordered associativity further constrain the quotient. The self-dual positive-indicator normalization imposes the established square-channel congruences modulo four. These conditions are necessary and cover every categorification, whether pivotal or not; a signed solution is not itself a categorification.

The rank-six proof of the general construction and the dimension-order obstruction [17] are rank-independent. Here their enumeration engine is extended to seven basis elements. It enumerates the independent cyclic/dual-reflection coefficient orbits, prunes only exact associativity equations with checked integer bounds, and constructs *every* signed quotient. For each quotient the checker verifies either its adjoint parity, its gauge equivalence with the original ring, or the failure of every admissible spherical character by codegree and dimension-order tests. In particular the dimension-order test requires the central-induction dimension δ/f_χ to lie in the $\mathbb{Z}$-span of the signed simple dimensions, as required by [11, 17].

Proposition 3 (Verified exhaustive branches)**.** *The oriented-dimension-order criterion excludes four, fifty-one, sixty-one, and one hundred thirty previously surviving rank-seven rings of exact multiplicities one through four, respectively. For* R7_1_5*,* R7_1_23*,* R7_1_33*, and* R7_1_38*, the complete signed enumerations have respectively* $2, 4, 4, 8$ *associative quotient tables. Every table in each of these 246 certified exhaustive searches is eliminated by necessary conditions for arbitrary complex fusion categories.*

Proof. The machine-readable proof tree for each of the 246 rings gives its full dimension-character factorization, every obstruction, the ordered signed-orbit domain, and one checked branch for every enumerated table. The independent verifier rebuilds the domains from the source tensor, repeats the bounded exhaustive enumeration, checks the *entire* list of surviving tables for equality to the recorded list, and recomputes each branch witness. The search used to discover these finite trees is logically separate from their exhaustive replay; timed-out or unfinished searches generate no negative decision. The enumeration returns the four displayed multiplicity-one branch counts and no uncovered branch for any of the 246 certificates. The rank-independent necessity of the search is the sphericalization theorem cited above. □

4 Further theoretical exclusions

The local triangular-prism proof of Liu–Palcoux–Ren–Vercleyen [6, Corollary 6.2, Theorem 6.4 and Theorem 7.2] excludes *R*7_2_185 and *R*7_3_635 over $\mathbb{C}$. The first is exactly their $\mathcal{F}_{210}$. The second is a consequence of the same local proof, as follows. In the source basis of dimensions $(1, 5, 5, 5, 6, 7, 7)$, their tensors agree outside the cube on the two dimension-seven objects. Both have this strictly positive integral dimension character, so any categorification is pseudo-unitary and has the canonical spherical structure. The cited proof uses the squares of two dimension-five objects k, l, selected columns of dimensions one and five, their dualities and categorical dimensions. The zero constraints are read from the squares of those selected objects. Thus none of the differing coefficients on the dimension-seven cube enters either local subsystem or its linking equation. Both rings give precisely the inconsistent polynomial system in the proof of [6, Theorem 7.2]. The supplementary checker verifies all these tensor equalities and the positive dimension character; it does not rerun the published Gröbner-basis calculation.

For *R*7_3_900 and *R*7_3_991, a different short argument gives a general exclusion. Their dimension type is $(1, 1, 1, 1, 2, 4, 4)$ and their Frobenius–Perron dimension is $40 = 2^3 \cdot 5$. Any categorification would be solvable [7, Theorem 1.6], hence weakly group-theoretical and unitarizable

[8, Theorem 5.20]. For a unitary categorification, the positive pth-root limit of [9, Corollary 3.4] requires, for every subset S of basis labels and $N_{ij}^l > 0$,

$$\frac{\sum_{x\in S} N_{lk}^x d_x}{d_l} \le \frac{\sum_{m\in T} N_{jk}^m d_m}{d_j}, \qquad T = \{m : N_{jk}^m > 0,\ N_{im}^x > 0 \text{ for some } x \in S\}.$$

Taking S to be the objects of dimension at most two, the exact witnesses in the supplement give $2 \le 1$ for each ring. The first witness is $(i, j, k, l) = (1, 1, 2, 1)$; the second is $(3, 4, 4, 3)$, using zero-based census labels. This unitary obstruction is a general one here because the dimension theorem forces unitarizability.

Proposition 4 (A half-partition criterion)**.** *Let R be an integral commutative fusion ring of dimension $4pq$, for distinct odd primes p, q. Write $a_E = 4pq/f_E$ for its formal codegrees. Suppose an odd-prime-power $a_E > 1$ occurs, and every submultiset of the a_E excluding the unit and summing to $2pq$ leaves an odd-prime-power entry in its complement. Then any categorification is weakly group-theoretical and has the Frobenius property.*

Proof. Integrality gives the canonical positive spherical structure. By [11, Theorem 2.13], the induced unit $I(1)$ in the center has distinct multiplicity-one summands of dimensions a_E. An odd-prime-power summand implies a nontrivial symmetric subcategory of the center [7, Corollary 7.2]. If this provides no nontrivial Tannakian subcategory, it must be sVec. For its fermion f, [11, Proposition 2.8] gives $s[I(1)] = 4pq[I(1)]$. The fermion has twist -1, so it is absent from $I(1)$, whose twist is trivial. Its row therefore gives $\sum_E \epsilon_E a_E = 0$, with monodromy signs $\epsilon_E = \pm 1$ and $\epsilon_{\mathbf{1}} = 1$. The negative signs form a half-partition. By hypothesis an odd-prime-power summand centralizes f. Its slightly degenerate centralizer consequently contains a nontrivial Tannakian subcategory by [7, Proposition 7.4].

Write this subcategory as $\mathrm{Rep}(G)$. Its de-equivariantization has neutral dimension $(4pq/|G|)^2$, hence $|G|$ divides $4pq$. If p or q divides $|G|$, the neutral category has at most two prime divisors and is solvable. Extension and equivariantization reconstruct the center; Morita invariance and subcategory closure then make the original category weakly group-theoretical [7, Theorem 1.6 and Proposition 4.1]. Otherwise $|G|$ is two or four and supplies $\mathrm{Rep}(C_2)$. The original category is a C_2-extension or C_2-equivariantization of a category of dimension $2pq$ [7, Propositions 2.9–2.10]. That category is integral: in the equivariantization case its simple dimensions are rational quotients of integral dimensions by orbit sizes and projective representation degrees, hence integers. The integral squarefree-dimension theorem [7, Theorem 9.2] makes it group-theoretical. The Frobenius property follows from [7, Theorem 1.5]. □

For *R7_3_905*, the exact positive dimension vector is $(1, 6, 6, 1, 1, 1, 8)$, and the formal codegrees are $(140, 20, 20, 7, 4, 4, 4)$. Thus the induced-unit dimensions are $(1, 7, 7, 20, 35, 35, 35)$. The only half-partition excluding the unit consists of two dimension-35 summands, leaving both dimension-seven summands. The proposition applies, but 6 does not divide 140. This excludes every complex categorification. The checker verifies the entire formal-codegree polynomial and all 64 nonunit subsets exactly.

Finally, *R7_4_2727* has dimension type $(1, 1, 1, 3, 6, 6, 6)$, dimension 120, and formal codegrees $(120, 60, 60, 24, 4, 3, 3)$. The induced unit has simple summands of dimension two, so the center contains a nontrivial symmetric subcategory. The invertible group has odd order three; [12, Lemma 2.5] therefore gives a nontrivial Tannakian subcategory, and [12, Lemma 3.1] makes any categorification weakly group-theoretical and hence unitarizable. The exact support inequality above gives $4 \le 2$ for $(i, j, k, l) = (3, 4, 3, 3)$ and $S = \{6\}$, a contradiction.

4.1 A double-equivariantization obstruction in dimension 52

The ring $R7_2_75$ has dimension type $(1^4, 4^3)$, invertible group V_4, and trivial universal grading. Every invertible fixes every dimension-four simple. These are exact tensor properties checked by the supplement. Suppose $\mathcal{C}$ categorifies it. Dimension $52 = 4 \cdot 13$ makes $\mathcal{C}$ solvable [7, Theorem 1.6]. The last step of its prime cyclic extension/equivariantization construction [7, Proposition 4.4] cannot be an extension, since $\mathcal{C}_{\mathrm{ad}} = \mathcal{C}$. Its character group must embed into V_4, so the last step is $\mathcal{C} = \mathcal{D}^{C_2}$, with $\mathrm{FPdim}(\mathcal{D}) = 26$. Each possible nontrivial character fixes the three dimension-four simples and pairs the four invertibles. The equivariant-simple orbit/stabilizer formula [1, Proposition 4.26] therefore forces $\mathcal{D}$ to have type $(1^2, 2^6)$ and rank eight. A fixed simple of the action has two lifts exchanged by the character; a free orbit has one lift of twice the dimension. There are no higher-dimensional projective representations of C_2.

The category $\mathcal{D}$ also has trivial universal grading. Equal dimensions of faithful grading components [21, Proposition 8.20] make the grading-group order divide 26. Orders 13 and 26 exceed the rank. Order two would give a neutral category of prime dimension 13, necessarily pointed and of rank 13 [21, Corollary 8.30], also impossible. Applying the same solvable last-step argument gives

$$\mathcal{D} = \mathcal{P}^{C_2}, \qquad \mathcal{P} = \mathrm{Vec}^{\omega}_{C_{13}}\,.$$

The inner C_2 must act nontrivially on C_{13}, hence by inversion; a trivial object action would give 26 invertible simples in $\mathcal{D}$.

The canonical central $\mathrm{Rep}(C_2)$ in $\mathcal{D}$ is unique over its underlying two invertibles. Indeed, the kernel of $\mathrm{Inv}(Z(\mathcal{D})) \to \mathrm{Inv}(\mathcal{D})$ is the group of tensor automorphisms of $\mathrm{Id}_{\mathcal{D}}$. Such an automorphism is trivial on every summand of $Y \otimes Y^*$, hence on $\mathcal{D}_{\mathrm{ad}} = \mathcal{D}$. Two lifts of the same invertible differ by this kernel. The outer C_2 action thus preserves the central $\mathrm{Rep}(C_2)$. Equivariantizing it gives a central Tannakian $\mathrm{Rep}(G)$ with fully faithful underlying embedding into $\mathcal{C}$, where $1 \to C_2 \to G \to C_2 \to 1$. Its symmetric fibre functor is the composite of the two forgetful functors. Therefore G is C_4 or V_4; its four characters exhaust the invertibles of $\mathcal{C}$, forcing $G = V_4$. De-equivariantization in stages now gives $\mathcal{C} = (\mathrm{Vec}^{\omega}_{C_{13}})^{V_4}$, with the normal C_2 acting by inversion [7, Section 2.6 and Proposition 2.10] [1, Theorem 4.18(i) and Proposition 4.19]. These are central Tannakian constructions; no braiding of $\mathcal{C}$ is assumed.

The object action $V_4 \to \mathrm{Aut}(C_{13}) = C_{12}$ has image of exponent at most two and contains inversion. Every nonidentity object consequently has orbit size two and stabilizer C_2. Each of the six nonidentity orbits contributes two dimension-two simples, while the unit contributes the four characters of V_4. This forces type $(1^4, 2^{12})$, contradicting $(1^4, 4^3)$. Thus $R7_2_75$ has no complex fusion categorification, without any pivotal or unitary assumption. The proof does not require ω to be trivial.

4.2 An order-sixteen orbit obstruction in dimension 96

The same method excludes $R7_2_252$ and $R7_2_315$. Both have dimension type $(1, 1, 2, 3, 3, 6, 6)$, invertible group C_2, and trivial universal grading. Their nontrivial invertible h fixes the dimension-two and dimension-six simples and exchanges the two dimension-three simples. Solvability and the last-step argument force $\mathcal{C} = \mathcal{D}^{C_2}$. The orbit/stabilizer formula then gives $\mathcal{D}$ dimension 48 and type $(1^3, 3^5)$.

The universal grading of $\mathcal{D}$ is trivial. Otherwise its group has order dividing 48, is solvable by Burnside's theorem, and has a prime cyclic quotient of order two or three. Equal component dimensions [21, Proposition 8.20] make both impossible. For a C_2 grading all three invertibles lie

in the neutral component, requiring $24 = 3 + 9n$. For a C_3 grading their group maps trivially or isomorphically to the grading group, requiring $16 = 3 + 9n$ or $16 = 1 + 9n$. Thus solvability forces $\mathcal{D} = \mathcal{E}^{C_3}$ with $\mathrm{FPdim}(\mathcal{E}) = 16$. Its unit accounts for all three invertibles of $\mathcal{D}$. A dimension-three simple of $\mathcal{D}$ comes from either a free orbit of three invertibles of $\mathcal{E}$ or a fixed dimension-three simple, the latter giving three such simples in $\mathcal{D}$. Counting the five dimension-three simples leaves only types (1^{16}) or $(1^7, 3)$ for $\mathcal{E}$. The second is impossible: its seven invertibles would fix its unique noninvertible Y, so YY^* would contain all seven, leaving dimension two that cannot be filled by copies of Y. Consequently $\mathcal{E} = \mathrm{Vec}_H^\omega$ for a group H of order sixteen.

As in the preceding proof, $\mathcal{D}_{\mathrm{ad}} = \mathcal{D}$ makes the canonical central $\mathrm{Rep}(C_3)$ unique over its underlying invertibles. The outer C_2 action preserves it. Equivariantization in stages gives $\mathcal{C} = (\mathrm{Vec}_H^\omega)^G$, where $1 \to C_3 \to G \to C_2 \to 1$ and the central $\mathrm{Rep}(G)$ embeds fully faithfully into $\mathcal{C}$. The cyclic group of order six would give six invertibles; hence $G = S_3$. The identity orbit contributes dimensions $1, 1, 2$, exhausting the small simples. Any other fixed orbit has projective stabilizer S_3, whose irreducible projective degrees are at most $\sqrt{6}$, and would give an extra small simple. An orbit of size two has stabilizer C_3 and would give extra dimension-two simples. An orbit of size three gives two dimension-three simples, and a free orbit gives one dimension-six simple. Thus the action on H must have orbit sizes

$$(1, 3, 6, 6).$$

Its normal C_3 acts without fixed points outside the identity; a transposition fixes exactly two elements of H.

Here is a purely group-theoretic contradiction. For a fixed-point-free order-three automorphism α of a group of order sixteen, every characteristic subgroup has order congruent to one modulo three. Thus the center has order four or sixteen. If it has order four, it is V_4, and the noncyclic quotient $H/Z(H)$ is also V_4. The commutator is an α-equivariant alternating map $\bigwedge^2 V_4 \to Z(H)$. Its source has trivial C_3 action, whereas its target has no nonzero fixed vector. It vanishes, a contradiction. So H is abelian. The characteristic subgroups $2H$ and $4H$ exclude C_{16}, $C_8 \times C_2$, and $C_4 \times C_2^2$, leaving $H = C_4^2$ or C_2^4.

Every involution of either remaining group fixes at least four elements. For C_2^4, $(S - I)^2 = 0$ gives $\dim \ker(S - I) \geq 2$. For C_4^2, if S reduces to the identity modulo two, it fixes all four elements of $2H$. Otherwise a lifted basis change gives $S = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ with a, b, d odd and c even. From $S^2 = I$ one obtains $bc = 0$ and $b(a + d) = 0$, hence $c = 0$ and $d = -a$ modulo four. Each of the four choices of x has the unique fixed-vector completion $y = -b^{-1}(a - 1)x$; the second equation follows from $a^2 = 1$. This contradicts the required two fixed points and excludes both rings over $\mathbb{C}$. The supplement checks the exact tensor premises and also provides an independent finite corroboration of this theoretical lemma.

4.3 Prime descent and finite orbit obstructions

The following families use the same mechanism. Solvability supplies a last cyclic-prime extension or equivariantization [7, Theorem 1.6 and Proposition 4.4]. When the adjoint subcategory is the whole category, an extension is impossible. The remaining equivariantization is constrained by the complete action of its character subgroup on the source simples. In an equivariantization, a simple above an orbit of Y has dimension

$$[G : G_Y] \dim(\pi) \, \mathrm{FPdim}(Y),$$

where π is an irreducible projective representation of G_Y [5, Section 2]. The source-bound checks below include all dimension products, adjoint closures, and character-subgroup permutations.

Thus descent is a categorical argument applied to checked hypotheses, rather than an identification by dimension type.

Proposition 5. *None of the twelve rings*

$$R7_2_50,\ R7_2_70,\ R7_2_113,\ R7_2_213,\ R7_2_273,\ R7_2_280,$$
$$R7_3_386,\ R7_3_398,\ R7_3_478,\ R7_3_966,\ R7_4_358,\ R7_4_963$$

is the Grothendieck ring of a complex fusion category.

Proof. We organize the argument by the descent obstruction.

Dimensions 40 *and* 136. For $R7_2_70$, $R7_2_213$, $R7_4_358$ the type is $(1^4, 2, d, d)$, with $(p, d) = (5, 4)$ or $(17, 8)$ and global dimension $8p$. The universal grading is trivial, and every invertible fixes each noninvertible. Prime descent therefore gives $\mathcal{C} = \mathcal{D}^{C_2}$ with $\mathcal{D}$ of type $(1^4, (d/2)^4)$ and dimension $4p$. If $\mathcal{D}$ admitted a C_2 grading, the outer involution would preserve a nonzero character of its 2-primary grading quotient; this would descend to a nontrivial grading of $\mathcal{C}$. A C_p grading is also impossible: at $p = 17$ its number of components exceeds rank eight; at $p = 5$ its neutral category is pointed of dimension four and its other four components have rank one. The latter force trivial pointed associator, while neither $\mathrm{BrPic}(\mathrm{Vec}_{C_4})$ nor $\mathrm{BrPic}(\mathrm{Vec}_{C_2^2})$ has an element of order five. Hence $\mathcal{D} = \mathcal{E}^{C_2}$.

Integrality of the descended dimensions and the strong Frobenius property [7] leave $\mathcal{E}$ of type $(1^2, 2^{(p-1)/2})$ and dimension $2p$. For clarity, at dimension ten the alternatives are pointed, $(1^6, 2)$, and $(1^2, 2^2)$: the middle case violates subcategory integrality, while the pointed case gives the forbidden invariant C_2 grading. At dimension thirty-four the dimension partitions are $(1^2, 2^{8-4b}, 4^b)$, $b = 0, 1, 2$; divisibility excludes $b > 0$. The same grading test gives $\mathcal{E}_{\mathrm{ad}} = \mathcal{E}$, so $\mathcal{E} = (\mathrm{Vec}^{\omega}_{C_p})^{C_2}$ with inversion on C_p.

The kernel of $\mathrm{Inv}\, Z(\mathcal{E}) \to \mathrm{Inv}\, \mathcal{E}$ is $\mathrm{Aut}_{\otimes}(\mathrm{Id}_{\mathcal{E}}) = \mathrm{Hom}(U(\mathcal{E}), \mathbb{C}^{\times}) = 1$. Consequently its central $\mathrm{Rep}(C_2)$ is unique and is preserved by the successive outer actions. Combining the equivariantizations [1, 7] gives $\mathcal{C} = (\mathrm{Vec}^{\omega}_{C_p})^H$, where $|H| = 8$ and $H_{\mathrm{ab}} = C_2^2$. Its action on C_p has cyclic image of order at most two. At $p = 17$, orbit size times projective stabilizer degree is at most four, contradicting the dimension-eight simples. At $p = 5$, choose a strictly H-invariant μ_p-valued associator by averaging. The vanishings $H^2(C_p, \mathbb{C}^{\times}) = 0$ and $H^2(H, \widehat{C_p}) = 0$ remove the action tensorators and their coherence multipliers. Thus the stabilizer representations are ordinary; their degrees are at most two, and nontrivial orbits have abelian stabilizer of order four. All dimensions are at most two, contradicting $d = 4$.

Dimension 112. For $R7_3_386$, $R7_3_966$, type $(1^2, 2^3, 7^2)$ and trivial universal grading force a C_2 descent of type $(1^7, 7)$ and dimension 56. Its adjoint category is again the whole category. Its last prime step is therefore a C_7 equivariantization of $\mathrm{Vec}^{\omega}_{C_2^3}$, with a Singer seven-cycle on the nonzero elements. The unique central C_7 survives the outer involution, so the combined group has order fourteen. Its representation subcategory has type $(1^2, 2^3)$ and hence is $\mathrm{Rep}(D_{14})$. The resulting action would conjugate the Singer cycle to its inverse. Its normalizer in $\mathrm{GL}_3(\mathbb{F}_2)$ is 7:3, which contains no such involution. This is impossible.

Dimension 96. For $R7_2_113$, $R7_2_273$, the character involution fixes the dimension-two and both dimension-six objects and exchanges the two dimension-three objects. These are exactly the premises of the dimension-96 proof above; its $(1, 3, 6, 6)$ orbit contradiction applies. For $R7_2_50$, $R7_2_280$, the involution instead exchanges both the dimension-three pair and the dimension-six pair. Descent has type $(1^3, 3, 6)$ and dimension 48, with trivial universal grading. Its next descent is a C_3 equivariantization of a dimension-16 category $\mathcal{E}$ of type $(1^4, 2^3)$, with the

three nontrivial invertibles and the three dimension-two objects each cyclically permuted. The invertibles form $V = C_2^2$ and fix the dimension-two objects. Hence $\mathcal{E}_{\mathrm{ad}} = \mathrm{Vec}_V^{\omega}$, with a faithful V grading whose three nonneutral components have rank one. A rank-one pointed module implies $[\omega] = 0$ by its module pentagon. There are only two rank-one left module categories over Vec_V, since $H^2(V, \mathbb{C}^\times) = C_2$. Distinct invertible grading components here must have inequivalent left module categories: an equivalence, tensored with the inverse of one component, would identify a nonneutral rank-one component with the rank-four regular module. Three such components cannot fit into two classes.

Dimension 54. For *R*7_3_398, *R*7_3_478, type $(1^2, 2^4, 6)$ forces descent of type $(1^9, 3^2)$ and dimension 27. Its pointed adjoint has group A of order nine; the other two C_3 grading components have rank one, so its associator is trivial. The outer involution inverts both A and the grading C_3. Its restricted tensorator has class $\eta \in H^2(A, \mathbb{C}^\times)$. Inversion acts trivially on this odd-order group, while order-two coherence gives $2\eta = 0$; hence $\eta = 0$. Its Brauer–Picard class restricts to $-I$ on $A \oplus \widehat{A}$, which is central in the orthogonal group [32, Corollary 1.2]. It therefore cannot conjugate a nontrivial order-three component class to its inverse, as the grading action requires.

Dimension 310. For *R*7_4_963, type $(1^2, 2^2, 10^3)$ has dimension $310 = 2 \cdot 5 \cdot 31$. The *pqr* theorem [7, Theorem 9.2] makes any categorification $\mathcal{C}(G, \omega, H, \psi)$ with $|G| = 310$. The complete closed-subring scan gives dimensions $1, 2, 10, 310$, so $|H|$ belongs to this list. For $|H| = 1, 2$ the double-coset formula [2, Theorem 5.1] cannot supply dimension ten. For $|H| = 10$, the representation subcategory forces $H = D_{10}$. The three outside dimension-ten simples force trivial double-coset intersections and outside cosets of size one hundred. Thus G would be a Frobenius group with kernel C_{31} and complement D_{10}; a faithful complement action in the cyclic group $\mathrm{Aut}(C_{31})$ is impossible. For $H = G$, the category is $\mathrm{Rep}(G)$. Its normal Sylow 31 subgroup has a split complement D_{10}, whose cyclic action on C_{31} has image at most two. The orbit/stabilizer formula then bounds all representation degrees by two, again contradicting ten. □

4.4 Four additional exhaustive signed exclusions

The complete oriented search excludes *R*7_3_1036, *R*7_3_1044, *R*7_4_2510, *R*7_4_2706. Their complete lists have respectively $8, 24, 4, 48$ signed associative tables. Every possible dimension character fails either nonvanishing, duality, or formal-codegree norm divisibility. The supplement regenerates the lists and rechecks each branch, including under optimized Python. These are general exclusions: sphericalization doubles both global dimension and the relevant formal codegrees, leaving the necessary quotient unchanged. No pivotal structure on the original category is assumed.

4.5 A relative-trace obstruction without signed enumeration

Proposition 6 (Relative-trace criterion)**.** *Let R be a commutative fusion ring with self-dual basis, and put $B = \sum_{i,j,k} (N_{ij}^k)^2$. Suppose $L \in \mathbb{Z}_{>0}$ and f are formal codegrees of R, and set $F = \mathbb{Q}(f)$. If every nonzero totally positive element of $L\mathcal{O}_F \cap f\mathcal{O}_F$ has field trace greater than B, then R has no categorification over $\mathbb{C}$.*

Proof. Sphericalize a hypothetical category and take its quotient by $\varepsilon = -1$. Both lifts of a self-dual simple remain self-dual [11, Section 5.1(a)]; hence the signed multiplication matrices are symmetric and commute. Writing them as M_i, their Casimir satisfies

$$\mathrm{Tr}\Big(\sum_i M_i^2\Big) = \sum_{i,j,k} (T_{ij}^k)^2 \le B.$$

Let d_i be the lift dimensions, $K = \mathbb{Q}(d_i)$, and $\delta = \sum_i d_i^2$. Every embedding of K gives a distinct dimension character and contributes its conjugate of δ as a Casimir eigenvalue. This counts $[K : \mathbb{Q}]$ characters even when some eigenvalues coincide. All Casimir eigenvalues are positive, so $\mathrm{Tr}_{K/\mathbb{Q}}(\delta) \leq B$.

The sphericalization has dimension 2δ and formal codegrees $2L, 2f$. The induced-unit theorem gives algebraic integers $\delta/L, \delta/f$ in its dimension field K [11, Theorem 2.13 and Corollaries 2.14–2.15]. Thus $F \subseteq K$. The element $t = \mathrm{Tr}_{K/F}(\delta)$ is totally positive and lies in $L\mathcal{O}_F \cap f\mathcal{O}_F$. Its trace is at most B, a contradiction. No equality $K = \mathbb{Q}(\delta)$ is assumed. □

Proposition 7. *The rings R7_4_1417 and R7_4_1420 admit no complex fusion-category categorification.*

Proof. Both rings have formal codegrees 20 and $f = 510 + 70\sqrt{51}$, whose minimal polynomial is $z^2 - 1020z + 10200$. Here $\mathcal{O}_F = \mathbb{Z}[\sqrt{51}]$. Write $t = 20(x + y\sqrt{51})$ in the intersection ideal. Then

$$\frac{t}{f} = (x - 7y) + \left(y - \frac{7x}{51}\right)\sqrt{51}.$$

Integrality forces $51 \mid x$, while total positivity forces $x > 0$. Consequently $\mathrm{Tr}_{F/\mathbb{Q}}(t) = 40x \geq 2040$. The exact Casimir trace bounds are 1086 and 1070, respectively, so Proposition 6 applies. The source-bound checker recomputes both characteristic polynomials, trace bounds, and the displayed quotient identity using exact arithmetic. □

4.6 A cyclic-stabilizer divisibility obstruction

Proposition 8. *Let $H \cong C_p$, with p prime, be a subgroup of invertible objects of a complex fusion category. Suppose that X is a simple object fixed on the left by H, Y is simple and fixed on the right, and Z is simple and fixed on both sides. If $\mathrm{FPdim}(Z)/p$ is not an algebraic integer, then p divides N_{XY}^{Z}.*

Proof. The one-element orbit of X is a rank-one module category over the pointed subcategory Vec_H^{ω}. Choosing the isomorphisms $hX \simeq X$ shows that ω is a coboundary, so $A = \bigoplus_{h \in H} h$ has its connected separable algebra structure. The obstruction to giving X a left A-module structure lies in $H^2(C_p, \mathbb{C}^{\times}) = 0$; similarly Y admits a right A-module structure. Thus $X \otimes Y$ is an A-bimodule, with the actions on its two factors.

The A-bimodule block supported on Z is described by a twisted group algebra of $C_p \times C_p$. Its projective irreducible representations have dimension one or p: after rescaling the two generators, their operators satisfy $U^p = V^p = 1$ and $UV = \zeta VU$ with $\zeta^p = 1$. If $\zeta = 1$ the algebra is commutative; otherwise ζ is primitive and every irreducible representation has dimension p.

For a simple bimodule with underlying object mZ, the Frobenius–Perron dimension in the Morita dual category is $m\,\mathrm{FPdim}(Z)/\mathrm{FPdim}(A) = m\,\mathrm{FPdim}(Z)/p$; see also the support and dimension argument in [19, Lemmas 6.1–6.2]. Algebraic integrality rules out $m = 1$. Every simple bimodule in this block therefore has underlying object pZ. Decomposing the bimodule $X \otimes Y$ proves the required divisibility. □

Proposition 9. *The rings R7_4_1212, R7_4_1213, R7_4_1438, R7_4_1440, and R7_4_2731 have no categorification over $\mathbb{C}$.*

Proof. In every row of the following table, the census element $g = X_1$ has order p and fixes X, Y, Z on both sides. The quotient norm is computed in the indicated quadratic dimension field.

ring	p	(X, Y, Z)	N_{XY}^{Z}	FPdim(Z)	Norm(FPdim$(Z)/p$)
R7_4_1212, *R7_4_1213*	2	(X_2, X_2, X_3)	1	$8 + 3\sqrt{5}$	19/4
R7_4_1438, *R7_4_1440*	2	(X_2, X_2, X_2)	1	$6 + 3\sqrt{5}$	$-9/4$
R7_4_2731	3	(X_3, X_3, X_3)	2	$5 + \sqrt{13}$	4/3

A nonintegral norm rules out algebraic integrality. Every displayed multiplicity violates Proposition 8. The source verifier checks the subgroup action and all equations for the exact positive dimension characters. No pivotal, spherical, or unitary hypothesis is used. □

Appendix A supplies an independent signed spectral proof for the first two rings, including a complete exact finite certificate.

4.7 Dimension-block obstructions and their scope

Proposition 10. *The rings R7_3_695, R7_3_783, and R7_3_1015 have no pivotal categorification over* $\mathbb{C}$.

Proof. For a spherical fusion category whose global dimension D generates a real quadratic field of discriminant f, the nontrivial-character case of Schopieray's dimension-block theorem [27, Theorem 5.1] gives

$$\left(\frac{1}{D} - \frac{1}{D'}\right)^2 \leq \frac{1}{e^2 f} \leq \frac{1}{f},$$

where e is a positive integer. The relevant source dimensions and exact values are

ring	D	f	$(1/D - 1/D')^2$
R7_3_695	$420 + 70\sqrt{35}$	140	1/35
R7_3_783, *R7_3_1015*	$120 + 30\sqrt{15}$	60	1/15.

Both conjugates violate the necessary inequality in each case.

It remains to exclude other dimension characters. Their global dimensions must be eigenvalues of the regular Casimir. For *R7_3_695*, the other roots are $5, 7$, and Norm$(D) = 4900$; the norms of $5/D$ and $7/D$ are $1/196, 1/100$. For the other two rings the rational roots are $6, 15$, and Norm$(D) = 900$; the corresponding norms are $1/25, 1/4$. Thus none of the rational alternatives satisfies formal-codegree divisibility [11, Corollary 2.14]. In the noncommutative case it is harmless to include the Casimir root from the matrix block in this necessary superset.

Finally every pivotal dimension character for these rings is spherical. All objects of *R7_3_695* are self-dual. In the other two rings, the three invertibles form C_3, a self-dual singleton E satisfies $gE = E$, and the other three simples form one C_3-orbit. Nonvanishing of $\dim(E)$ forces $\dim(g) = 1$; the three orbit dimensions are equal, so every simple and its dual have the same dimension. □

This proposition alone is a pivotal obstruction, not a general one. The full sphericalization doubles the global dimension, dividing the left-hand side of the displayed inequality by four; the strict violations then become equalities. No general exclusion is inferred solely from this dimension-block argument. The companion verifier keeps this scope explicit and checks all numerical and tensor hypotheses exactly.

4.8 A signed three-parameter argument forcing sphericality

Proposition 11. *The rings R7_3_783 and R7_3_1015 have no categorification over* $\mathbb{C}$.

Proof. We show that every hypothetical categorification would be spherical; Proposition 10 then gives a contradiction. The two source rings have basis $1, g, g^2, E, \rho_0, \rho_1, \rho_2$, where $g^3 = 1$, $E^* = E$, $\rho_0^* = \rho_0$, and $\rho_i = g^i\rho_0$. Put $G = 1 + g + g^2$ and $S = \rho_0 + \rho_1 + \rho_2$. Their rules are

$$gE = Eg = E, \qquad \rho_0 g = g^r \rho_0,$$
$$E^2 = G + 3E + S, \qquad E\rho_i = \rho_i E = E + 2S, \qquad \rho_0^2 = 1 + 2E + 2S,$$

where $r = -1$ for *R7_3_783* and $r = 1$ for *R7_3_1015*. These relations generate all products. In census order one takes $g = X_1$, $\rho_0 = X_6$, and respectively $E = X_4, X_5$; the verifier compares all 343 structure constants.

Sphericalize a hypothetical category and pass to the signed quotient. Choose the lift of g with $g^3 = 1$, choose positive-second-indicator lifts of E and ρ_0, and define the other lifts by $\rho_i = g^i\rho_0$. Since gE has the same underlying simple as E, its signed coefficient is ± 1. Its cube is one, so $gE = E$. The same argument gives $Eg = E$ and $\rho_0 g = g^r\rho_0$.

The coefficient of ρ_0 in E^2 is one: its original multiplicity is one and the positive-indicator square-channel congruence is modulo four. Left multiplication by g fixes E^2, forcing the other two orbit coefficients also to be one. Frobenius reciprocity determines the three invertible coefficients. Hence

$$E^2 = G + bE + S, \qquad b \in \{-1, 3\}.$$

The element $E\rho_0$ is fixed by left multiplication by g, and $E\rho_i = E\rho_0$. Its support and Frobenius reciprocity give

$$E\rho_i = E + aS, \qquad \rho_0^2 = 1 + aE + \sum_{i=0}^{2} c_i\rho_i,$$

where $a \in \{-2, 2\}$ by the square-channel congruence, and $|c_i| \le 2$. Put $c = \sum_i c_i \le 6$. Multiplying the second identity by G gives $S\rho_0 = G + 3aE + cS$. Comparing the coefficients of S in $E^2\rho_0 = E(E\rho_0)$ yields

$$ba + c = 3a^2.$$

The choices $(a, b) = (-2, -1), (-2, 3), (2, -1)$ require respectively $c = 10, 18, 14$, all impossible. Thus $a = 2$, $b = 3$, and $c = 6$, forcing every $c_i = 2$.

All displayed signed products are therefore the original positive products. The g-relations generate the remaining products, so the chosen seven lifts are closed under tensor product and duality. They form a spherical fusion subcategory of the sphericalization. The forgetful functor restricts to a tensor equivalence with the original category, which is therefore spherical, as required. □

The signed argument is purely theoretical; the companion verification checks only identification with the source tensors and elementary arithmetic. In particular it does not presume that based-ring symmetries furnish a pivotal structure. The remaining ring in Proposition 10, *R7_3_695*, is excluded generally by Proposition 15 below.

4.9 The full induced-unit compression bound

The following refinement keeps all simple constituents of the induced unit in the local compression argument of [27, Lemma 4.1 and proof of Theorem 5.1]. It requires no assertion about a dimensional grading for a general spherical category.

Proposition 12 (Compression over all constituents of $I(\mathbf{1})$)**.** *Let $\mathcal{C}$ be a spherical fusion category, $D = \dim(\mathcal{C})$, $K = \mathbb{Q}(\dim X : X \in \mathrm{Irr}(\mathcal{C}))$, and $G = \mathrm{Gal}(K/\mathbb{Q})$. For an irreducible representation E of $K_0(\mathcal{C}) \otimes \mathbb{C}$, write f_E for its formal codegree. Then $f_E \in K$, and for every character χ of G,*

$$\sum_E \left| \sum_{\sigma \in G} \chi(\sigma)\sigma(f_E^{-1}) \right|^2 \leq \frac{|G|}{\mathfrak{f}(\chi)}. \tag{1}$$

There is no factor $\dim E$ in the outer sum.

Proof. Let A_E be the simple constituent of $I(\mathbf{1})$ attached to E. The induction theorem [11, Theorem 2.13] gives $\dim A_E = D/f_E$, $\theta_{A_E} = 1$, and distinct E give distinct A_E. Since the forgetful image of every object of the center has dimension in K, this also proves $f_E \in K$. Write $s = S/D$ for the normalized center matrix and P for orthogonal projection onto its twist-one subspace. For the dimension character and its Galois conjugates let A_σ denote the corresponding constituents of $I(\mathbf{1})$, and set $w_\chi = \sum_{\sigma \in G} \chi(\sigma) e_{A_\sigma}$. The Galois compatibility of the modular representation gives $s_{A_\sigma, Y} = \sigma(\dim Y/D)$, and w_χ is fixed by the unipotent subgroup and has torus character χ. The local compression estimate for $SL_2(\mathbb{Z}/N)$, combined over prime powers, gives $\|Psw_\chi\|^2 \leq \|w_\chi\|^2/\mathfrak{f}(\chi) = |G|/\mathfrak{f}(\chi)$. Its A_E coordinate is exactly the inner sum in (1). Retaining these coordinates proves the claim. The multiplicity $[I(\mathbf{1}) : A_E] = \dim E$ does not repeat a coordinate of the modular representation. □

Corollary 13 (Quadratic Casimir obstruction)**.** *Let R be a fusion ring, let N_i be its regular fusion matrices, and let $B = \sum_i N_i N_i^{\mathsf{T}}$. Fix a real quadratic field L of fundamental discriminant q. List the irreducible quadratic factors of $\det(xI - B)$ whose roots generate L as*

$$\prod_j (x^2 - t_j x + n_j)^{m_j}, \qquad Q_L(R) := q \sum_j m_j \frac{t_j^2 - 4n_j}{n_j^2}.$$

Assume that at least one such factor occurs. If R has a spherical categorification, then $Q_L(R) \leq 1$. More precisely, if K is its dimension field, then $L \subseteq K$ and $[K : L]Q_L(R) \leq 1$.

Proof. The Casimir eigenvalue on a simple representation E is $\lambda_E = (\dim E) f_E$, with regular multiplicity $(\dim E)^2$. In particular $L \subseteq K$. Put $e' = [K : L]$ and pull back to $\mathrm{Gal}(K/\mathbb{Q})$ the nontrivial quadratic character of L, whose conductor is q. If $\lambda_E \in L$, its contribution to (1) is $e'^2(\dim E)^2(\lambda_E^{-1} - \overline{\lambda_E}^{-1})^2$. Regular multiplicities therefore account for the squares of representation degrees, including when R is noncommutative. Each displayed factor contributes $2e'^2 m_j(t_j^2 - 4n_j)/n_j^2$. The upper bound is $2e'/q$, which gives the asserted inequality after cancellation. □

Corollary 14 (Boundary constraint for an arbitrary categorification)**.** *If R has any fusion-category categorification, then $Q_L(R) \leq 4$. If $Q_L(R) = 4$, its sphericalization has dimension field exactly L, and every formal codegree of its $\varepsilon = -1$ sector is rational. In particular its actual spherical global dimension is rational.*

Proof. The sphericalization has two lifts of every simple object and a central invertible ε of order two and dimension -1. The $\varepsilon = 1$ quotient of its fusion algebra is the original fusion algebra; in this sector its Casimir is $2B$. Thus the factors supplied by this sector contribute $Q_L(R)/4$ to the preceding proof. Its dimension field contains L, so $[K : L]Q_L(R)/4 \leq 1$. At equality $Q_L(R) = 4$, necessarily $K = L$, and the known sector already exhausts the squared-norm bound. All other $I(\mathbf{1})$ coordinates must vanish under the nontrivial Fourier character of L, so their formal codegrees are rational. The spherical dimension character belongs to the $\varepsilon = -1$ sector, and its formal codegree is the global dimension. □

4.10 A parity contradiction at the compression boundary

Proposition 15. *The ring R7_3_695 has no categorification over* $\mathbb{C}$.

Proof. Its source basis is self-dual and its Casimir polynomial is

$$P(x) = (x-7)^3(x-5)^2(x^2-840x+4900).$$

The irrational roots are $D = 420 + 70\sqrt{35}$ and its conjugate. Their field $L = \mathbb{Q}(\sqrt{35})$ has discriminant 140, and

$$Q_L(R) = 140\frac{840^2 - 4 \cdot 4900}{4900^2} = 4.$$

For a hypothetical categorification, the boundary case of Corollary 14 therefore says that its sphericalization has dimension field $K = L$ and all formal codegrees in the signed sector are rational.

Both lifts of each source simple are self-dual, so the signed quotient is commutative with self-dual basis. Its seven codegrees are eigenvalues of its integer Casimir matrix B_T. They are thus positive integers. Let $d = (d_i)$ be its spherical dimension character and $\delta = \sum_i d_i^2$. The field generated by the d_i is $K = L$, so d and its quadratic conjugate are distinct. Both have the same rational integer codegree δ. Formal-codegree divisibility gives

$$\frac{\delta}{D} = \frac{\delta}{70}(6 - \sqrt{35}) \in \mathcal{O}_L.$$

Since $\mathcal{O}_L = \mathbb{Z}[\sqrt{35}]$, we obtain $70 \mid \delta$.

The signed coefficients agree with the source coefficients modulo two. Hence $B_T \equiv B_N \pmod 2$ and

$$\det(xI - B_T) \equiv P(x) \equiv x^2(x+1)^5 \pmod 2.$$

Exactly two signed codegrees are even, namely the two occurrences of δ. Write the other five, which are odd, as $h_1, \ldots, h_5$. Induction integrality gives $\delta/h_i \in \mathbb{Z}$, and the reciprocal-codegree identity yields

$$\frac{2}{\delta} + \sum_{i=1}^{5} \frac{1}{h_i} = 1, \qquad \sum_{i=1}^{5} \frac{\delta}{h_i} = \delta - 2.$$

Since δ is even and every h_i is odd, $\delta/h_i \equiv \delta \pmod 4$. The last equality would therefore give $5\delta \equiv \delta - 2 \pmod 4$, a contradiction. □

This argument uses the individual dimension-character field throughout. It does not infer a nontrivial grading from rational spherical global dimension. Its exact source verifier checks the Casimir polynomial, compression equality, inverse formula, and both even residue classes modulo four.

4.11 A dimension-150 central boson

Proposition 16. *The census ring $R_{7,2,160}$ has no categorification over $\mathbb{C}$.*

Proof. Its basis is $1, g, v, X_1, \dots, X_4$, with dimensions $(1, 1, 2, 6, 6, 6, 6)$ and total dimension 150. All basis objects are self-dual, the adjoint subring is the whole ring, and

$$g^2 = 1, \quad gv = v, \quad gX_i = X_i, \quad v^2 = 1 + g + v, \quad vX_i = 2X_i.$$

Suppose that $\mathcal{C}$ realizes this ring. It is integral and pseudo-unitary [21]. Its center is integral and modular. The modular formal-codegree formula implies that every central simple dimension d divides 150: the rational number $150^2/d^2$ is an algebraic integer, so the rational number $150/d$ is an integer.

Write $F : \mathcal{Z}(\mathcal{C}) \to \mathcal{C}$ for the forgetful functor and I for its right adjoint. The formal codegrees are $150, 3, 2, 25, 25, 25, 25$. Thus $I(1)$ has seven distinct constituents, each of multiplicity one, with dimensions $1, 50, 75, 6, 6, 6, 6$ [11, Theorem 2.13]. Each restriction contains 1 once. The formula $FI(W) = \bigoplus_{S \in \mathrm{Irr}(\mathcal{C})} SWS^*$ gives

$$\langle I(1), I(g) \rangle = 5, \qquad \langle I(1), I(v) \rangle = 9, \qquad \langle I(g), I(g) \rangle = 7.$$

For a dimension-six constituent Y of $I(1)$, write $F(Y) = 1 + bg + cv$; then $b + 2c = 5$, so b is positive and odd. The four such constituents use at least four of the total five copies of g. Consequently all four have $b = 1, c = 2$. If $b_{50}, b_{75}, c_{50}, c_{75}$ are the corresponding coefficients for the other two nontrivial constituents, the induction pairings give $b_{50} + b_{75} = c_{50} + c_{75} = 1$. Reduction of their dimensions modulo six gives $1 + b_{50} + 2c_{50} \equiv 50 \pmod 6$, hence $(b_{50}, c_{50}, b_{75}, c_{75}) = (1, 0, 0, 1)$. It follows that $I(g)$ shares precisely five multiplicity-one constituents with $I(1)$, of total dimension 74. Its squared norm is seven, and its dimension is 150, so its other two constituents are distinct, of multiplicity one, and have dimensions summing to 76. The only pair of positive divisors of 150 with this sum is $(1, 75)$. A central invertible object $\widetilde{g}$ over g is therefore forced.

The actual fusion subcategory $\mathcal{D} = \langle 1, g, v \rangle$ has a rank-one module category generated by any X_i. This gives a fiber functor; semisimple Hopf reconstruction in dimension six identifies $\mathcal{D}$ with $\mathrm{Rep}(S_3)$ [20]. Its universal grading is trivial, so the half-braiding on g is unique: two choices differ by a tensor automorphism of the identity. Its restriction is therefore the symmetric sign object, with self-braiding $+1$. The universal grading of $\mathcal{C}$ is also trivial, so a central invertible over 1 is trivial. Hence $\widetilde{g}^2 = 1$, and $\langle \widetilde{g} \rangle$ is a central $\mathrm{Rep}(C_2)$.

De-equivariantization yields $\mathcal{C} = \mathcal{E}^{C_2}$ with $\mathrm{FPdim}(\mathcal{E}) = 75$. The equivariantization orbit formula [5] and the sign action give $\mathcal{E}$ the type $1^3, 3^8$: each sign-fixed dimension-two or dimension-six source simple comes from a free orbit of two objects of dimension one or three. For a dimension-three simple Z of $\mathcal{E}$, its invertible stabilizer has order one or three. Order one would leave dimension eight in ZZ^* to be supplied by dimension-three objects, which is impossible. Thus ZZ^* contains all three invertibles and two dimension-three constituents. The adjoint subcategory therefore has dimension $3 + 9k$ with $k > 0$; divisibility by 75 forces it to equal $\mathcal{E}$.

By the two-prime theorem, $\mathcal{E}$ is solvable [7, Theorem 1.6]. Its last prime step cannot be an extension, since its universal grading is trivial; it is an equivariantization by C_3, since its invertible group is C_3. The resulting central $\mathrm{Rep}(C_3)$ has these invertibles as its underlying objects. Their half-braidings are unique, so the outer C_2 action preserves this central subcategory. Equivariantizing it embeds a central symmetric category of dimension six into $\mathcal{Z}(\mathcal{C})$: its half-braidings with objects of $\mathcal{E}$ descend to equivariant objects. Forgetting equivariance gives a fiber functor, so this category is $\mathrm{Rep}(H)$ for a group of order six. It has exactly two invertibles, because

it contains the canonical $\mathrm{Rep}(C_2)$ and embeds in $\mathcal{C}$. Thus $H = S_3$, and its image is precisely $\langle 1, g, v\rangle$. Direct de-equivariantization has dimension 25, hence is pointed [21, 19]. We obtain

$$\mathcal{C} = (\mathrm{Vec}_A^{\omega})^{S_3}, \qquad |A| = 25.$$

For a nonidentity element of A in an orbit of length $\ell < 6$, a simple lift has dimension $\ell \dim \pi$, where π is an irreducible projective representation of a stabilizer of order $6/\ell$. It is at most $\sqrt{6\ell} < 6$. Such a lift cannot belong to $\langle 1, g, v\rangle$, whose objects lie over the identity element of A, and every other simple of $\mathcal{C}$ has dimension six. Thus S_3 acts freely on $A \setminus \{1\}$. If $A = C_{25}$, its automorphism group has order twenty and the order-three subgroup acts trivially. If $A = C_5^2$, every involution acting without nonzero fixed vectors is $-I$, since its minimal polynomial divides $(t-1)(t+1)$. All three involutions would therefore have the same image, making their products, the order-three elements, act trivially. Both cases contradict freeness. □

The exact certificate checks the complete source tensor, the adjoint subring, formal codegrees, the three induction pairings, and the small integer and finite-group premises. It does not enumerate a full induction matrix.

5 Three fixed simples and an anti-invariant module

The following argument uses only semisimple module categories and the ordinary fusion rules. In particular, its conclusions are general.

Proposition 17 (Anti-invariant module representation)**.** *Suppose a commutative fusion ring is categorified, an invertible simple g has order two, and exactly f simples $F_1, \dots, F_f$ are fixed by g. There are integral $f \times f$ matrices T_i satisfying*

$$T_0 = I, \qquad T_{i^*} = T_i^{\mathsf{t}}, \qquad T_i T_j = \sum_k N_{ij}^k T_k,$$

$$|(T_i)_{ab}| \le N_{iF_a}^{F_b}, \qquad (T_i)_{ab} \equiv N_{iF_a}^{F_b} \pmod 2.$$

Each T_i is normal, and its spectrum is contained in that of N_i.

Proof. A fixed simple gives a rank-one module category over the pointed subcategory generated by g, trivializing its associator. Thus $A = 1 \oplus g$ is a connected separable algebra. Each fixed simple has two right A-module structures, since $H^2(C_2, \mathbb{C}^\times) = 0$; a free orbit gives one simple module. Twisting by the algebra automorphism that is -1 on g exchanges the two structures over each F_a and fixes the free-orbit modules. This involution commutes with the left category action. On its minus eigenspace, use the basis $F_a^+ - F_a^-$. The two nonnegative action coefficients have sum $N_{iF_a}^{F_b}$, so their difference has the asserted bounds and parity. Module reciprocity gives transpose duality. Commutativity then gives normality. The squarefree characteristic polynomial of N_i annihilates b_i in the semisimple complexified fusion ring, and hence annihilates T_i. □

Theorem 18. *None of*

$$R7_2_116,\ R7_2_261,\ R7_3_364,\ R7_3_938,$$
$$R7_3_970,\ R7_4_2644,\ R7_4_2683$$

is categorifiable by a complex fusion category.

Proof. All indices below are source indices, starting with $X_0 = 1$.

For $R7_2_116$ and $R7_2_261$, take respectively $(g, X) = (X_1, X_2)$ and (X_3, X_1), with fixed ordering (X_4, X_5, X_6). The bound matrix for T_X is

$$B = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

The source spectra lie respectively in $\{0, \pm 1, 5\}$ and $\{0, \pm 1, \pm i, 5\}$. The maximum absolute row sum is at most four, excluding eigenvalue five. Normality implies operator norm at most one. Its first row nevertheless contains two entries of absolute value one, a contradiction.

For $R7_3_364$ and $R7_3_938$, take respectively $g = X_1, X_3$, again with fixed ordering (X_4, X_5, X_6). The self-dual simple X_4 has bound matrix and squarefree source polynomial

$$B = \begin{pmatrix} 3 & 1 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 2 \end{pmatrix}, \qquad p(x) = x(x^2 - 8x - 8)(x^2 - 2x - 2).$$

Write its symmetric anti-invariant matrix as

$$T = \begin{pmatrix} a & b & c \\ b & d & e \\ c & e & f \end{pmatrix}, \qquad a, b, d \text{ odd}, \quad c, e, f \text{ even}.$$

Its integral characteristic polynomial has degree three. Thus its spectrum is three zeros, $\{0, 4 \pm 2\sqrt{6}\}$, or $\{0, 1 \pm \sqrt{3}\}$. The first option contradicts symmetry and odd entries. The second has $\operatorname{tr}(T^2) = 80$, whereas the bounds give $\operatorname{tr}(T^2) \leq 32$. In the last option the trace, trace-square, and determinant are $2, 8, 0$. The trace-square and parity force $c = e = 0$, $|a| = |b| = |d| = 1$, and $|f| = 2$. The determinant forces $ad = b^2 = 1$, so $a = d$. Consequently $a + d + f \in \{-4, 0, 4\}$, contradicting trace two.

For $R7_3_970$, take $g = X_3$ and fixed ordering (X_4, X_5, X_6). The bound matrix for T_1 is

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 1 \end{pmatrix},$$

and its source spectrum lies in $\{0, \pm 1, \pm i, 9\}$. The row-sum bound three excludes nine. Normality and parity therefore give $T_1 = T_2 = D = \operatorname{diag}(0, \varepsilon, \eta)$, where $\varepsilon, \eta = \pm 1$. The respective bounds on T_5, T_6 are

$$\begin{pmatrix} 0 & 2 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 0 \\ 2 & 0 & 2 \end{pmatrix}.$$

Commutation with D kills both off-diagonal pairs. The remaining diagonal supports are disjoint, so $T_5 T_6 = 0$. The source product $X_5 X_6 = 2X_1 + 2X_2$ instead gives $T_5 T_6 = 4D \neq 0$.

For $R7_4_2644$, take $g = X_3$, fixed ordering (X_4, X_5, X_6), and test simple X_1. Its bounds and polynomial are

$$\begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{pmatrix}, \qquad (x - 1)(x + 1)(x^2 + 1)(x^2 - 14x + 1).$$

The row-sum bound seven excludes $7 + 4\sqrt{3}$, and rationality of the characteristic polynomial excludes its conjugate. Normality and parity imply $T_1 = T_2 = D$, diagonal with entries ± 1. The first column of T_5 is zero by the source bounds. Thus $(T_4T_5)_{00} = 0$, whereas $X_4X_5 = 2X_1 + 2X_2$ forces this entry to be ± 4.

Finally, for *R7_4_2683*, take $g = X_5$ and fixed ordering (X_1, X_2, X_6). The same bounds and polynomial, now for X_3, give $T_3 = T_4 = D$, diagonal with entries ± 1. The first row of T_1 is zero, while the source product $X_1^2 = 2X_3 + 2X_4$ forces $(T_1^2)_{00} = 4D_{00} = \pm 4$. This completes all seven contradictions. □

The accompanying source verifier reconstructs the involutions, fixed simples, duality, bound matrices, spectral polynomials, and displayed products directly from the pinned census. The proof does not require a search through associators or sphericalization signs.

6 Morita-dual Gram identities

The same module category gives stronger necessary conditions than its fusion-ring representation alone. These conditions use only Frobenius–Perron dimensions and semisimplicity.

Proposition 19 (Two evaluation Gram identities)**.** *Let $A = 1 \oplus g$ be the involution algebra above, $\mathcal{M} = \mathcal{C}_A$, and $\mathcal{D} = {}_A\mathcal{C}_A$. Write M_i for the action matrices of the source simples on the simple right A-modules, and normalize their positive dimension vector d by $d_A = 1$. The dimensions of the simple objects of $\mathcal{D}$ are obtained from the unique nonnegative integral column decomposition of $I + M_g$: there are two singleton columns at each fixed point of the involutive permutation M_g, and one pair-column at each transposition. A column v gives dimension $\sum_a v_a d_a$.*

For every simple module m, there must also be nonnegative integral columns v_Y, one for each simple Y of $\mathcal{D}$, such that

$$G_m := \sum_i (M_i)_{mm} M_i = \sum_Y v_Y v_Y^{\mathsf{t}}, \qquad \sum_a (v_Y)_a d_a = \mathrm{FPdim}(Y) d_m. \tag{†}$$

The column of the unit of $\mathcal{D}$ is e_m.

Proof. The commuting actions of $\mathcal{C}$ and $\mathcal{D}$ make $\mathcal{M}$ a Morita equivalence. Tensor–Hom adjunction for evaluation at m gives the Gram matrix in (†): the internal end of m in $\mathcal{C}$ is $\bigoplus_i (M_i)_{mm} X_i$, and its action on $\mathcal{M}$ is the composite of evaluation and its adjoint. The columns are the multiplicities of the simples in $m \otimes_A Y$. Positivity of Frobenius–Perron dimensions gives the dimension identity, with $d_m = \mathrm{FPdim}_{\mathcal{C}}(m)/2$. At $m = A$, the internal end is $A = 1 \oplus g$, giving $I + M_g$. Its zero entries separate fixed points and transpositions. A fixed point has diagonal two, so it supplies two unit columns; a transposition supplies a single column with two ones. This proves uniqueness of the decomposition. Every simple bimodule has nonzero underlying right module, so none is omitted. □

Theorem 20. *The seven rings*

$$R7_2_11,\ R7_2_205,\ R7_2_275,\ R7_2_287,$$
$$R7_3_387,\ R7_4_2473,\ R7_3_1055$$

have no categorification over $\mathbb{C}$.

Proof. The four $\sqrt{2}$ cases. The exact anti-module equations have one sign-gauge orbit for each of R7_2_11, R7_2_205, R7_2_275, and R7_2_287. This small statement is checked independently by the complete source-bound anti-module enumerator. It also follows by using the normality and spectral restrictions of the preceding proposition: for a noninvertible generator, the squared Frobenius norm is at most 20, whereas the conjugate pair $2 \pm 2\sqrt{2}$ contributes 24. The remaining pair $\pm\sqrt{2}$, or $\pm i\sqrt{2}$, forces its even entries to vanish and its four unit couplings to have absolute value one; the fusion relations determine the remaining matrices up to changing the signs of the three anti-module basis vectors.

Put $\delta = 1 + \sqrt{2}$. The resulting right-module dimensions are $1^4, \delta^4$, and the dual dimensions are $1^8, \delta^8$. For a recorded module simple of dimension δ, the full internal-end Gram matrix restricted to the four δ coordinates is

$$G_\delta = \begin{pmatrix} 4 & 2 & 2 & 2 \\ 2 & 4 & 2 & 2 \\ 2 & 2 & 4 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}.$$

Let $S(Q) = \operatorname{tr}(Q) - 2\sum_{i<j} Q_{ij}$ on this block. A dual invertible acts by a column whose δ coordinates sum to one, so its contribution to S is one. A dual simple of dimension δ has column dimension $\delta^2 = 1 + 2\delta$; hence its δ coordinates are nonnegative integers summing to two. Its contribution to S is either zero or four. All contributions are therefore nonnegative, whereas $S(G_\delta) = 16 - 24 = -8$. This contradiction excludes all four cases without an optimizer or any pivotality assumption.

The integral case R7_3_387. The bound and spectrum of T_2 give $T_2 = \operatorname{diag}(0, \varepsilon, \eta)$ with $\varepsilon, \eta = \pm 1$, exactly as in the dimension-240 case of Theorem 18. Commutation kills the possible off-diagonal entries of T_5, T_6, leaving disjoint diagonal supports. The product $X_5X_6 = 2X_2 + 2X_3$ gives $T_3 = -T_2$. Since $gX_2 = X_3$, the last two entries of T_g are -1. As g fixes X_4, X_5, X_6, their last two diagonal entries vanish. The source spectrum and bounds force $T_4 = \operatorname{diag}(-1, 0, 0)$; then $X_4^2 = 1 + g + X_4$ forces $T_g = \operatorname{diag}(1, -1, -1)$. Thus $T_5 = T_6 = 0$, and precisely the four choices of (ε, η) remain. In all four, the module dimensions are

$$(1, 9, 1, 1, 3, 3, 3, 3),$$

and the dual dimensions determined by $I + M_g$ are $(1^6, 6^2, 9^2)$. At the first dimension-three module, the full matrix G_m has rows $0, 2, 3$ identical, with common diagonal entry one. They force a unique column

$$v = (1, 1, 1, 1, 1, 1, 0, 0)^{\mathsf{t}}.$$

Its dimension is eighteen, so it corresponds to one of the dual simples of dimension six. After subtracting vv^{t}, rows $0, 2, 3$ vanish. The six invertible columns each have exactly one entry one among coordinates $4, 5, 6, 7$, and vanish at coordinate one. For the remaining three columns, coordinate one's squared entries sum to twelve. The only expression of twelve as three nonnegative integer squares is $4 + 4 + 4$. Hence each entry there is two. The remaining dual dimension-six column has no other nonzero entry. The two dimension-nine columns have last-four-coordinate vectors w, z with

$$\sum w_a = \sum z_a = 3, \qquad w + z = (1, 1, 2, 2).$$

The $(4, 5)$ Gram entry, after the forced column is removed, is one. It gives $w_4 = w_5 \in \{0, 1\}$. The $(4, 6)$ and $(4, 7)$ entries are also one, forcing $w_6 = w_7 = 1$. Thus $\sum w_a = 2w_4 + 2 \in \{2, 4\}$, contradicting three. The entries used are identical in all four anti-invariant branches.

Two remaining finite Gram certificates. For R7_4_2473 and R7_3_1055, the complete anti-module systems have respectively seven and one sign-gauge orbits. For each orbit the verifier

reconstructs the entire finite column domain of one internal-end Gram matrix, including its exact FP dimension constraint, and checks an integral separating vector y with $y^T A \geq 0$ and $y^T b < 0$. All eight systems are impossible. The public verifier reopens the census, regenerates all anti-module orbits and all Gram columns, and checks the frozen inequalities with exact arithmetic; it never invokes an optimizer. Together with the four cases above and the dimension-240 argument, this gives seven further general complex exclusions.

□

6.1 A positive-semidefinite column refinement

The rank-one columns in Proposition 19 satisfy an additional exact constraint. This sharpens the finite Gram systems without introducing categorical assumptions.

Proposition 21 (Rank-one positive-semidefinite constraint)**.** *Let Q be a nonzero rational symmetric positive-semidefinite matrix, let I index a maximal independent set of columns, and put $P = Q_{I,I}$. A column v can occur in a decomposition $Q = \sum_j v_j v_j^{\mathsf{t}}$ only if*

$$v = Q_{:,I} P^{-1} v_I, \qquad v_I^{\mathsf{t}} P^{-1} v_I \leq 1. \tag{‡}$$

These two conditions are equivalent to $Q - vv^{\mathsf{t}} \succeq 0$. A nonpositive-semidefinite Q admits no such column decomposition.

Proof. Every summand has range contained in that of Q, giving the first identity. Since Q is positive semidefinite and its selected columns are independent, P is positive definite. Set $B = Q_{:,I} P^{-1}$. The rank identity $Q = BPB^{\mathsf{t}}$ and $v = Bv_I$ reduce the condition to $P - v_I v_I^{\mathsf{t}} \succeq 0$. Conjugating by $P^{-1/2}$ gives $I - ww^{\mathsf{t}} \succeq 0$, where $w = P^{-1/2} v_I$, exactly the second inequality. These are rational identities and inequalities: a verifier can clear denominators and use positive principal minors, without numerical eigenvalues. □

Theorem 22. *The four rings R7_4_776, R7_4_927, R7_4_2126, and R7_3_909 have no categorification over $\mathbb{C}$.*

Proof. Proposition 17 gives a finite exact anti-module system for each source tensor. In the stated order, these systems have 55, 20, 12, and 27 sign-gauge orbits. For each orbit, take a recorded simple module m and remove the unit column from its evaluation Gram matrix: $Q = G_m - e_m e_m^{\mathsf{t}}$. The nonnegative integral columns already satisfy the dimension and coordinate bounds of Proposition 19. Retain only those satisfying (‡). The remaining Gram and dimension-class count equations form a finite system $Ax = b$, $x \geq 0$.

All 114 systems have frozen integral separating vectors satisfying $y^{\mathsf{t}} A \geq 0$ and $y^{\mathsf{t}} b < 0$. The public verifier reopens the source census, regenerates every anti-module orbit, reconstructs each full column domain, applies the rational range and positive-semidefinite tests, and checks these inequalities. It uses no optimizer and no numerical spectral comparison. Thus every necessary module branch is impossible, proving the four general exclusions. □

7 Two Morita transformations and an A_4 obstruction

The following argument uses actual module categories of a hypothetical categorification. Its finite checks only verify the source products, eight alternating forms, and the elementary ring normalization.

Theorem 23. *The rings R7_2_236, R7_4_2167, and R7_4_2716 have no categorification over* $\mathbb{C}$.

Proof. Their dimension vectors are $(1,1,1,3,d,d,d)$, with $d^2 = 4 + md$ and $m = 6, 12, 12$, respectively. The invertibles form $C_3 = \langle g \rangle$, fixing the dimension-three simple X and permuting the other three simples freely. The source products give $X^2 = 1 + g + g^2 + 2X$.

The first Morita transformation. The fixed simple X trivializes the pointed associator, so the regular algebra $A = 1 + g + g^2$ exists. Since $H^2(C_3, \mathbb{C}^\times) = 0$, X has both left and right A-module structures. Thus $X \otimes X$ is an A-bimodule. Its X-isotypic part is $2X$. A nontrivial projective class for $C_3 \times C_3$ has irreducible degree three, so this multiplicity-two part forces the class on X to be trivial. It gives nine simple bimodules with underlying object X, each of dual dimension one. The pointed orbit gives three more invertibles. The remaining orbit has cyclic stabilizer and gives three simples of dimension d.

Consequently $\mathcal{C}_A$ has dimensions $1^4, d$, and its Morita dual $\mathcal{D}$ has dimensions $1^{12}, d^3$. Already this excludes $m = 6$: the published [28, Proposition 3.3] requires $s \mid r$ for a category with two simple dimensions $1, d$ and irrational d satisfying $d^2 = s + rd$. Here $4 \nmid 6$.

For the two remaining cases, write K for the twelve invertibles of $\mathcal{D}$. They fix the unique d-dimensional object of $\mathcal{C}_A$; its rank-one module structure trivializes the whole pointed associator. Thus $\mathcal{D}_{\mathrm{pt}} \simeq \mathrm{Vec}_K$. The integral subcategory of $\mathcal{C}$ has rank four, and its A-module category has four simple objects on which K acts transitively. A stabilizer L has order three. If L were normal, the dual of Vec_K on this module category would have four double cosets, each contributing three characters of L, hence rank twelve instead of four. Therefore L is nonnormal. Its core is trivial because $|L| = 3$, so the action embeds K as an index-two subgroup of S_4; hence $K = A_4$.

For a noninvertible Y of $\mathcal{D}$, comparison of rational and irrational coefficients in $\mathrm{FPdim}(YY^*) = 4 + 12d$ shows that YY^* contains exactly four invertibles. Each occurs with multiplicity at most one by rigidity. Thus both stabilizers are the unique subgroup $H = V_4$ of A_4, and the three noninvertibles form one orbit.

The second Morita transformation. At least one of these three simples, say Y, is self-dual. The projective $H \times H$ stabilizer of Y has an alternating commutator form on $\mathbb{F}_2^4$. Duality exchanges the left and right factors and inverts the commutator. Its values are ± 1, so in matched bases the form is

$$\beta = \begin{pmatrix} bJ & C \\ C & bJ \end{pmatrix}, \qquad J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad C = C^{\mathsf{t}}.$$

This uses rigidity, with no pivotal assumption. The two algebra structures on $\bigoplus_{h \in H} h$ differ by the unique nontrivial class in $H^2(H, \mathbb{C}^\times)$, adding $\mathrm{diag}(J, J)$ to β. For the eight symmetric matrices C, the two ranks are $2, 4$ in some order, except for $C = 0$, when they are $0, 4$. The rank-zero alternative gives a simple bimodule of dimension $d/4$. Its monic minimum polynomial has constant coefficient $-1/4$, contradicting algebraic integrality. Hence one algebra choice gives rank two. The corresponding twisted group algebra has four irreducible representations of degree two, giving four simple bimodules over each Y, of dimension $t = d/2$. Transport by A_4 preserves the unique nontrivial H-cocycle class, so the same degree holds over all three noninvertibles.

The pointed Morita dual is again Vec_{A_4} for either algebra choice. For the untwisted algebra this is Fourier duality for $V_4 \rtimes C_3$. The other cocycle extends to A_4 through $Q_8 \rtimes C_3 \to A_4$, so an identity-on-objects tensor autoequivalence of the pointed subcategory identifies the two pointed duals. No extension of this autoequivalence to $\mathcal{D}$ is assumed. The resulting category $\mathcal{D}'$ therefore has dimensions $1^{12}, t^{12}$, with invertible group A_4 and $t^2 = 1 + 6t$. For every noninvertible Z, comparison of rational coefficients in $\mathrm{FPdim}(ZZ^*) = 1 + 6t$ gives a trivial stabilizer. Thus these twelve objects form a free A_4-orbit.

Normalization without a self-duality assumption. Choose any noninvertible ρ in $\mathcal{D}'$. Right multiplication is free as well, since duality turns a right stabilizer into a left stabilizer. Write

$$\rho g = \phi(g)\rho, \qquad \rho^2 = a + N\rho, \qquad N \in \mathbb{Z}_{\geq 0}[A_4], \quad \sum_g n_g = 6.$$

Rigidity gives a unique invertible summand a. Associativity gives

$$\phi^2 = \mathrm{Ad}_a, \quad \phi(a) = a, \quad Ns = \phi(s)N, \quad \phi(N) = N,$$

and then $\rho^* = a^{-1}\rho$ and $N^* = a^{-1}N$. Since $\mathrm{Aut}(A_4) = S_4$ and $Z(A_4) = 1$, write $\phi = \mathrm{Ad}_\sigma$; necessarily $a = \sigma^2$. Replacing ρ by $b\rho$ replaces (σ, a, N) by $(b\sigma, (b\sigma)^2, bN)$. Choose $b = \sigma^{-1}$ when σ is even, and $b = \tau\sigma^{-1}$ otherwise, with τ a transposition. In either case $b \in A_4$, the new a is one, and the new ρ is self-dual. The two normalized actions are therefore the identity and conjugation by τ.

For the identity action, centrality and self-adjointness give

$$N = A\,1 + B \sum_{h \in V_4 \setminus \{1\}} h, \qquad (A, B) = (6, 0), (3, 1), (0, 2).$$

The two size-four three-cycle classes would have equal coefficients, so neither can occur in a sum of size six. The first choice contains the forbidden rank-two subcategory $\rho^2 = 1 + 6\rho$ [25]. For the other choices, N acts on the three-dimensional representation of A_4 by $A - B = \pm 2$. Taking ρ to act by a root of $x^2 \mp 2x - 1$ gives the formal codegrees

$$16 + 8\sqrt{2}, \qquad 16 - 8\sqrt{2}.$$

For the transposition action, τN is constant on the two A_4-conjugacy classes in the odd coset of S_4: the six transpositions and the six four-cycles. It must be one of their class sums. On the standard three-dimensional representation of S_4, these class sums act by 2 and -2, respectively. Taking $\rho = rT$, where T represents τ and $r^2 = 1 \pm 2r$, again gives the displayed formal codegrees. In both computations the Casimir is $12(1 + \rho^2)$ and its scalar equals the representation degree times its formal codegree.

Finally, the published [28, p. 1628, Lemma 3.1] states that every category with two simple Frobenius–Perron dimensions is Galois conjugate to a pseudo-unitary category, with the same based ring. In this conjugate, the central constituent A_E associated to an irreducible representation has dimension D/f_E, where $D = \mathrm{FPdim}(\mathcal{D}')$ [11, Theorem 2.13]. Its forgetful image has dimension in $\mathbb{Z} + \mathbb{Z}t$, so every f_E must belong to $\mathbb{Q}(t)$. Here $t = 3 + \sqrt{10}$, whereas $16 \pm 8\sqrt{2}$ do not belong to $\mathbb{Q}(\sqrt{10})$. This contradiction excludes both remaining source rings. □

For $R7_3_1022$, the same reduction has $m = 8$ and $t^2 = 1 + 4t$. Its normalized system retains the branch $N = \sum_{h \in V_4} h$, which passes the formal-codegree field condition. Thus this argument makes no categorifiability decision for that source ring.

8 Exact constructions and replay

The three formerly undecided multiplicity-one rings match AnyonWiki $\mathrm{FR}_3^{7,2}$, $\mathrm{FR}_{12}^{7,2}$, and $\mathrm{FR}_3^{7,4}$, respectively, by full based-ring isomorphisms. For each, we use the published category with code $(7, 1, d, i, 1, 0, 1)$ in the Vercleyen–Slingerland census [13]. The symbolic associators supplied

through TensorCategories.jl lie over $\mathbb{Q}(\zeta_8)$, $\mathbb{Q}(\zeta_{28})$, and $\mathbb{Q}(\zeta_{16})$, respectively. Independent rational-polynomial replay checks all 9,856, 45,061, and 9,856 admissible pentagon equations, all 585, 837, and 585 nonempty associator blocks for invertibility, and the unit and rigidity constraints. These checks establish complex fusion categories directly, with no pivotality assumption. The AnyonWiki category list additionally records unitary gauges. Only existence of a model is used: neither completeness nor uniqueness of the external category census is required. The supplementary evidence records the exact category codes, cyclotomic coefficients, source hashes and basis permutations.

The positive model catalogue constructs complete fusion tensors from the cyclic group, ordinary finite-group character products, the Tambara–Yamagami rule, $SU(2)_k$ truncation, its even parts, odd metaplectic rules, and the categorical $\mathbb{Z}/2$ zest of $SU(2)_6$ [29, 22, 24, 30, 31]. Character products for $\mathrm{Rep}(S_5)$, $\mathrm{Rep}(SL_2(\mathbb{F}_3))$, $\mathrm{Rep}(C_{13} \rtimes C_3)$ and $\mathrm{Rep}(C_{11} \rtimes C_5)$ are computed with exact characters in the bundled software. Two further complete model tensors come from the published noncommutative integral appendix of [18, Theorem 1.6 and Section 6.2]: the representation category of its dimension-24 Kac algebra and the group-theoretical category $\mathcal{C}(A_5, 1, A_4, 1)$. The published full tensors and their byte digest are pinned in the package. For each constructed or published model, the program validates the fusion axioms, then checks all 6! unit-preserving basis permutations against every compatible census tensor. A positive record carries its explicit permutation. Different categorical structures may share a based ring; the count is of rings, not categories.

Three additional standard constructions give full-tensor matches: $R7_3_602 = K_0(\mathrm{Rep}(A_6))$, $R7_2_218 = K_0(\mathrm{Rep}(C_{13} \rtimes_5 C_4))$, and $R7_2_313 = K_0(\mathrm{Adj}(\mathrm{SU}(3)_5))$. The first uses exact character products over $\mathbb{Q}(\sqrt{5})$; the second uses induction of characters of C_{13}, with the complement generator acting by exponent five. The third is the triality-zero subcategory of the standard unitary quantum-group category [10]; exact Weyl alternants over $\mathbb{Q}(\zeta_{24})$ reconstruct all its fusion coefficients. Each is checked at all 343 structure constants.

8.1 Two dimension-24 equivariantizations

Put $\mathcal{T} = \mathrm{TY}(C_3, \chi, 1/\sqrt{3})$, with $\chi(a, b) = \exp(2\pi i ab/3)$. Write m for its noninvertible simple and $|a| = 0$, $|m| = 1$ for its C_2 grading. Inversion $I(a) = -a$, $I(m) = m$ preserves χ and gives a coherent unitary C_2 action with canonical channel identifications as tensorators and identity square coherence [4, Section 4A and Proposition 4.8]. The associators are phases, permutations or the normalized Fourier matrix.

First take $\mathcal{E} = \mathcal{T} \boxtimes \mathrm{Vec}(C_2)$ and $F(X, e) = (I(X), e + |X|)$. Grading additivity and the trivial pointed associator give the tensor structure, and $F^2 = \mathrm{Id}$ as a tensor functor. Thus $\mathcal{E}^{C_2}$ is unitary. Its two fixed simple objects $(0, e)$ each have two equivariant lifts t_e^+, t_e^-; the other invertibles form orbits $P_e = \{(1, e), (2, e)\}$, and the two odd objects form $Z = \{(m, 0), (m, 1)\}$. In the ordered basis

$$(1, h, t, ht, P_0, P_1, Z) = (t_0^+, t_0^-, t_1^+, t_1^-, P_0, P_1, Z),$$

the invertible group is C_2^2, h fixes both P_e, t exchanges them, and all invertibles fix Z. The remaining products are

$$P_e P_f = t_{e+f}^+ + t_{e+f}^- + P_{e+f}, \qquad P_e Z = Z P_e = 2Z,$$
$$Z^2 = 1 + h + t + ht + 2P_0 + 2P_1.$$

Here and below subscripts are modulo two. Fixed output channels are exchanged in pairs and have one invariant vector of each sign; free orbit outputs have a unique equivariant lift. These rules agree with $R7_2_1$ under the model-to-census permutation $(0, 5, 1, 2, 4, 3, 6)$.

For the second construction, let z generate C_4 and use the normalized unitary cocycle

$$\omega(a,b,c) = i^{w(a,b,c)}, \qquad w(a,b,c) = a\left\lfloor \frac{b+c}{4} \right\rfloor, \quad 0 \le a,b,c < 4.$$

Inside $\mathcal{T} \boxtimes \mathrm{Vec}(C_4,\omega)$ take the full diagonal subcategory with simples (a, z^{2e}), (m, z) and (m, z^3). It is closed under tensor products and duality because TY parity equals the pointed degree modulo two. Define F by inversion in both factors. Its TY tensorator is that of I; on pointed degrees use $J(a,b) = i^{u(a,b)}$, in the direction $F(X) \otimes F(Y) \to F(X \otimes Y)$, where

$$u = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 3 & 2 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

Reducing arguments modulo four, direct substitution gives

$$u(b,c) - u(a+b,c) + u(a,b+c) - u(a,b) \equiv w(a,b,c) - w(-a,-b,-c),$$
$$u(a,b) + u(-a,-b) \equiv 0 \pmod 4.$$

The first is tensor-functor coherence; the second makes F^2 the identity tensor functor. Identity multiplication coherence therefore defines a genuine C_2 action. The supplement checks all 256 pointed cocycle equations, 64 tensorator equations and 16 square equations.

The equivariantization has the same simple-orbit pattern. The positive lift t of $(0, z^2)$ satisfies $t^2 = h$, since $J(2,2) = -1$; hence its invertibles form C_4. All other displayed products remain valid, with $t_1^+ = t$ and $t_1^- = t^3$. In Z^2, the degree-two pair of channels is exchanged with coefficients $1, 1$, and the degree-zero pair with coefficients $i, -i$. Both involutions have trace zero and eigenvalues $1, -1$, giving one copy of each fixed lift. The complete tensor matches $R7_2_195$ under $(0, 6, 1, 2, 5, 3, 4)$. All scalars and tensorators are unitary, so both constructions are unitary categories over $\mathbb{C}$. Besides comparing all 686 coefficients, the replay independently reconstructs them as C_2-invariant dimensions of equivariant Hom spaces, using $(\dim + \mathrm{tr}(F))/2$.

8.2 A quaternion equivariantization in dimension 48

Let $Q_8 = \{\pm 1, \pm e_1, \pm e_2, \pm e_3\}$ with $e_1e_2 = e_3$, $e_2e_3 = e_1$, $e_3e_1 = e_2$, and $e_a^2 = -1$. For $p \in S_3$, the rule

$$\rho_p(e_a) = \mathrm{sgn}(p) e_{p(a)}$$

is a group automorphism, and $\rho_p\rho_q = \rho_{pq}$. It gives a strict unitary action on $\mathrm{Vec}(Q_8)$, so $\mathrm{Vec}(Q_8)^{S_3}$ is a unitary fusion category of dimension 48. Each fixed central grade $1, -1$ carries the three ordinary irreducibles $1, \varepsilon, V$ of S_3; the six noncentral grades form one free orbit, giving a simple X of dimension six [5]. Label the first six simples (z, a), with $z \in C_2$ and $a \in \{1, \varepsilon, V\}$. Their products are those of $\mathrm{Rep}(S_3) \boxtimes \mathrm{Vec}(C_2)$ and $(z,a)X = X(z,a) = \dim(a)X$. In X^2, the six ordered pairs with product either central grade form a regular S_3-set; each noncentral grade has four ordered pairs. Consequently, in the order $(0,1), (0,\varepsilon), (0,V), (1,1), (1,\varepsilon), (1,V), X$,

$$X^2 = (0,1) + (0,\varepsilon) + 2(0,V) + (1,1) + (1,\varepsilon) + 2(1,V) + 4X.$$

This complete tensor matches $R7_4_2$ under $(0, 5, 4, 1, 2, 3, 6)$. The replay checks the group action, computes the central permutation characters and compares all 343 coefficients.

8.3 The twisted quaternion model from a Chern class

The remaining dimension-48 ring $R7_4_2019$ has a unitary group-theoretical model

$$\mathcal{C}(\Gamma, \Omega, S_3, 1), \qquad \Gamma = Q_8 \rtimes S_3, \qquad [\Omega] = c_2(V) - \inf c_2(R).$$

The action is the signed-axis action above. To construct the class, view unit quaternions inside SU(2), and distinguish the scalar imaginary unit I from quaternion units i, j, k. Set

$$r = -\frac{1+i+j+k}{2}, \qquad s = I\frac{i-j}{\sqrt{2}}.$$

Then $r^3 = s^2 = 1$ and $srs = r^{-1}$; conjugation implements the signed axis cycle and transposition. Thus r, s define a unitary representation $R : S_3 \to U(2)$ with determinant the sign character, and $V(q, p) = qR(p)$ is a representation of Γ. The ordinary second Chern classes give a well-defined class in $H^4(B\Gamma, \mathbb{Z}) \cong H^3(\Gamma, U(1))$ [3]. Its restriction to S_3 is zero.

Put $z = -1 \in Q_8$ and $K = \langle z \rangle \times S_3$. If $x = c_1(L)$ for the sign line of $\langle z \rangle$ and $y = c_1(\det R)$, the splitting principle gives

$$[\Omega]|_K = c_2(L \otimes R) - c_2(R) = x^2 + xy.$$

Writing elements of K as (a, p) and parity as $\epsilon(p)$, this restriction has the explicit normalized representative

$$\omega_K((a,p),(b,q),(c,r)) = (-1)^{abc + a\epsilon(q)\epsilon(r)}.$$

Indeed the real logarithm $a(bc + \epsilon(q)\epsilon(r))/2$ has integer coboundary $ab(cd + \epsilon(r)\epsilon(s))$, representing $x^2 + xy$. The replay checks all 12^4 such identities exactly. Choose a unitary global cocycle representing $[\Omega]$. Extend a correcting normalized two-cochain from K to Γ to make its restriction exactly ω_K. Then $\Omega|_{S_3} = 1$. The group-theoretical category is the category of $\mathbb{C}[S_3]$-bimodules in $\mathrm{Vec}_\Gamma^\Omega$ [2, Section 3 and Theorem 5.1]. The unitary cocycle and standard positive separable subgroup algebra make it unitary. This proves existence without constructing a table of 48^3 cocycle values.

The central double cosets have stabilizer S_3. Over z, the multiplier is

$$\alpha_z(p,q) = \frac{\omega_K(z,p,q)\omega_K(p,q,z)}{\omega_K(p,z,q)} = (-1)^{\epsilon(p)\epsilon(q)}.$$

It has the one-dimensional projective representation $u(p) = I^{\epsilon(p)}$. Its lift t satisfies $t^2 = \varepsilon$: the central-degree tensorator is trivial and $u(p)^2$ is the sign character. Thus the six central simples are $1, \varepsilon, v, t, t\varepsilon, tv$, with the S_3 representation products and relation $t^2 = \varepsilon$. The noncentral double coset is free and gives one simple X of dimension six. Products $YX = XY = \mathrm{FPdim}(Y)X$ follow from support and dimension. Each central product fibre of X^2 is a free transitive S_3-set, so its projective monomial representation is projective regular, with irreducible multiplicities $1, 1, 2$ regardless of the cocycle phases. The noncentral fibres each have size four. Hence

$$X^2 = 1 + \varepsilon + 2v + t + t\varepsilon + 2tv + 4X.$$

In this order the model-to-census permutation is $(0, 6, 5, 1, 2, 3, 4)$. The supplement checks the representation relations, action, characteristic-class restriction, projective lift, and all 343 coefficients. The remaining 10 tensors are undecided.

8.4 The unitary D_{12} model

The ring $R7_2_240$ is the even part of the D_{12} subfactor, equivalently the local bosonic modularization of $PSU(2)_{20}$. Existence and the split-object fusion rules follow from [33, Theorem 4.14] and [34, Theorem 2.2 and Proposition 2.4]. The resulting category is unitary modular. In the model basis $f_0, f_2, f_4, f_6, f_8, P, Q$, bulk products are obtained by folding the $SU(2)_{20}$ rules under $f_{2t} = f_{2(10-t)}$ and setting $f_{10} = P + Q$. The remaining rules are

$$Pf_{2j} = \sum_{k=5-j}^{4} f_{2k} + \begin{cases} P & j \text{ even}, \\ Q & j \text{ odd}, \end{cases} \qquad Qf_{2j} \text{ interchanges } P, Q,$$

$$P^2 = Q + f_2 + f_6, \quad Q^2 = P + f_2 + f_6, \quad PQ = f_0 + f_4 + f_8, \quad P^* = Q.$$

The model-to-census permutation is $(0, 5, 3, 6, 4, 1, 2)$. It matches all 343 ordered structure constants; the separate associativity check has 2401 equations. The category is supplied by the cited construction, not inferred from those finite ring checks.

The complete package contains the pinned source, negative certificates, independent model tensors, stage summary and verification reports. For the frozen release, run `python3 verify.py` and `python3 -O verify.py`. The portable residual-stage verifier checks the later prime-descent and signed certificates, the D_{12} model, source identities and preservation of earlier decisions. From the repository root the two entry points are

```
python3 tools/verify_upper_residual_20261002b.py
python3 tools/verify_upper_residual_20261002c.py.
```

Run both also with `-O`. The subsequent exact verifiers are in

```
evidence/theoretical_20261016/rank7/
```

and are named

```
verify_signed_pair.py,
verify_C3_signed_general.py,
verify_cyclic_module.py,
verify_R695_general.py,
verify_three_fixed.py,
verify_three_fixed_gram.py,
verify_psd_gram.py,
verify_C3_A4_reduction.py,
verify_A4_quadratic.py,
verify_scoped_dimension_block.py.
```

Each accepts `-repo /path/to/repository`; the last makes no general decision by itself. The Gram verifiers use the frozen exact helpers in `gram_support/`; their replay does not invoke an optimizer. Run each normally and with `-O`. The archive dispatcher supplies both stages. Earlier full-certificate reports retain their own scopes and dates; preservation checks do not claim a fresh execution of all prior searches. Original discovery programs are retained separately and are not required to verify this frozen release.

AI disclosure. GPT-6 Astra assisted with research, symbolic exploration, code development, and drafting.

A Independent signed verification of two quadratic cases

A.1 A quadratic-field reduction for two signed quotients

Proposition 24. *For a hypothetical categorification of R7_4_1212 or R7_4_1213, choose positive-second-indicator lifts in its sphericalization and let d_i be their dimensions. Then*

$$K = \mathbb{Q}(d_0, \dots, d_6) = \mathbb{Q}(\sqrt{5}), \qquad \delta = \sum_i d_i^2 \in \{180, 360, 270 \pm 90\sqrt{5}, 450 \pm 90\sqrt{5}\}.$$

Proof. The two source Casimir polynomials are respectively

$$(z-90)(z-20)^2(z-4)(z-2)(z^2-900z+6480),$$
$$(z-90)(z-12)(z-10)(z-6)(z-2)(z^2-900z+6480).$$

Their traces are $B = 1036, 1020$, and the least common multiple of the rational formal codegrees is 180. As in Proposition 6, the signed multiplication matrices are commuting symmetric integer matrices. The formal-codegree field and divisibility theorems for the sphericalization [11, Theorem 2.13 and Corollaries 2.14–2.15] therefore give $\mathbb{Q}(\sqrt{5}) \subseteq K$ and $\beta = \delta/180 \in \mathcal{O}_K$. Every embedding of K is real, and β is totally positive.

The nontrivial invertible g fixes all five noninvertibles in the source. The square-channel congruence gives $T_{ii}^g = 1$, and Frobenius reciprocity gives $gX_i = X_i$ in the signed quotient. Thus $(1, -1, 0, \dots, 0)$ is a character of codegree two, distinct from every conjugate of the dimension character. If $e = [K : \mathbb{Q}]$, then e is even, $e \le 6$, and

$$180 \operatorname{Tr}_{K/\mathbb{Q}}(\beta) + 2 \le B.$$

Total positivity and the arithmetic–geometric mean inequality give $\operatorname{Tr}_{K/\mathbb{Q}}(\beta) \ge e$. Hence $e \neq 6$.

Suppose $e = 4$ and put $r = [\mathbb{Q}(\beta) : \mathbb{Q}] \mid 4$. If $r = 4$, its minimal polynomial is $z^4 - az^3 + bz^2 - cz + d$, with positive integer coefficients and

$$a \le 5, \qquad b \le \lfloor 3a^2/8 \rfloor, \qquad c \le \lfloor a^3/16 \rfloor, \qquad d \le \lfloor a^4/256 \rfloor.$$

These bounds follow from the trace inequality and Maclaurin's inequalities. Factoring every polynomial in this finite box and counting its positive roots exactly gives no irreducible totally positive quartic. The complete check is included in the verifier. If $r = 2$, a totally positive irreducible quadratic algebraic integer has trace at least three, so its trace from K is at least six, contradicting the same bound. Thus $r = 1$, and $4\delta + 2 \le B$ forces $\delta = 180$.

There would then be four signed Casimir eigenvalues 180, one of value two, and two further positive eigenvalues u, v. The reciprocal-codegree identity gives $90(u + v) = 43uv$. Their residual quadratic has integer coefficients, so it is

$$z^2 - 43kz + 90k \qquad (1 \le k \le 7).$$

Each of these seven quadratics is irreducible, while $(43k)^2/(90k) = 1849k/90$ is not integral. Their roots are therefore not d-numbers, contrary to the formal-codegree theorem [26]; doubling a codegree does not change the d-number property. This excludes $e = 4$.

Consequently $e = 2$ and $K = \mathbb{Q}(\sqrt{5})$. Write $\beta = (a + b\sqrt{5})/2$, with $a, b \in \mathbb{Z}$ of the same parity. Total positivity and the trace bound give $1 \le a \le 5$ and $a^2 > 5b^2$. The six possibilities are $(a, b) = (2, 0), (3, \pm 1), (4, 0), (5, \pm 1)$, proving the assertion. □

The exact arithmetic underlying this reduction is preserved in

`evidence/upper_residual_20261002c/scoped_research/QUADRATIC_REMAINDER.md`

at source commit `1592905dc573227ad03659a6370a9459d40db26c`. The present verifier independently replays its complete quartic box, seven quadratic d-number failures, and source Casimir checks.

A.2 Two quadratic signed cases

Proposition 25. *The fusion rings `R7_4_1212` and `R7_4_1213` have no categorification over* $\mathbb{C}$.

Proof. We use the sphericalization conditions and Proposition 24. Suppose that a categorification exists, and choose the lifts of positive second Frobenius–Schur indicator in its sphericalization. Write T for the signed quotient by the central sign object. Its structure constants satisfy

$$|T_{ij}^k| \le N_{ij}^k, \qquad T_{ij}^k \equiv N_{ij}^k \pmod 2, \qquad T_{ii}^j \equiv N_{ii}^j \pmod 4.$$

All basis elements are self-dual, so trace cyclicity and dual reflection make T_{ij}^k fully symmetric. The nontrivial invertible element g acts positively on all five noninvertible basis elements. The spherical dimension character $d = (1, 1, d_2, \dots, d_6)$ generates $K = \mathbb{Q}(\sqrt{5})$, all its coordinates are nonzero, and its codegree $\delta = \sum_i d_i^2$ belongs to

$$\{180, 360, 270 \pm 90\sqrt{5}, 450 \pm 90\sqrt{5}\}.$$

Every signed formal codegree lies in K, by applying the formal-codegree field theorem to the sphericalization and dividing its codegrees by two. It is also an algebraic integer, since it is an eigenvalue of the integer signed Casimir matrix $\sum_i M_i^2$. The characters d and $\bar{d}$ are distinct, including when δ is rational. The character $(1, -1, 0, \dots, 0)$ has codegree 2.

Consider multiplication by X_2 on the $g = 1$ summand, in the basis $(1 + g, X_2, \dots, X_6)$. Its matrix M is self-adjoint for $G = \mathrm{diag}(2, 1, 1, 1, 1, 1)$. Set

$$p(t) = \det(tI - M), \qquad q(t) = \det(tI - M_{\{1,\dots,5\},\{1,\dots,5\}}).$$

Here the first row of M is $(0, 1, 0, 0, 0, 0)$ and its first column is $(0, 2, 0, 0, 0, 0)^{\mathsf{t}}$. For an eigenvalue λ, let w_λ be the first diagonal entry of its G-orthogonal spectral projection. The resolvent identity gives

$$\frac{q(t)}{p(t)} = \sum_\lambda \frac{w_\lambda}{t - \lambda}.$$

Each common character of codegree h contributes $2/h$ to this weight. Indeed its eigenvector has coordinates $(1, \chi(X_2), \dots, \chi(X_6))$, whose squared G-norm is h. Consequently, if d_2 is simple, then $2p'(d_2) - \delta q(d_2) = 0$.

There are exactly 97 200 and 259 200 possible matrices M, respectively, under the displayed coefficient conditions. We enumerate only these matrices, without imposing the remaining associativity equations. For each matrix we test for common roots of p with

$$2p' - 180q, \quad 2p' - 360q, \quad (2p' - 270q)^2 - 40500q^2, \quad (2p' - 450q)^2 - 40500q^2.$$

The complete exact calculation, supplied by `verify_signed_pair.py` and `one_matrix.cpp`, is summarized below. The modular tests use the primes 1009 and 1013; all surviving gcds are

subsequently computed over $\mathbb{Q}$. Since p is monic, a characteristic-zero common factor cannot disappear upon reduction modulo either prime.

ring	matrices	first prime	both primes	simple roots surviving
`R7_4_1212`	97 200	996	372	0
`R7_4_1213`	259 200	1 888	288	0

More precisely, every common root remaining over $\mathbb{Q}$ lies in a repeated factor of p. All repeated factors are linear and have multiplicity two or three. Thus d_2 is rational, and the distinct characters d and $\bar{d}$ occupy its eigenspace. Their combined weight is

$$b = \frac{2}{\delta} + \frac{2}{\bar{\delta}} = \begin{cases} 1/45, & \delta = 180, \\ 1/30, & \delta = 270 \pm 90\sqrt{5}, \\ 1/90, & \delta = 360 \text{ or } 450 \pm 90\sqrt{5}. \end{cases}$$

For multiplicity two we must have $w_{d_2} = b$. For multiplicity three, the third character has codegree $h = 2/(w_{d_2} - b)$, which must be a positive integer. Exact rational projections exclude every multiplicity-three branch and every multiplicity-two branch except four matrices for `R7_4_1213`, all with $d_2 = -3$ and $\delta = 180$.

Each of these four matrices has

$$p(t) = (t-2)(t+3)^2(t^3 - 3t^2 - 5t + 3),$$
$$q(t) = (t+3)(t^4 - 2t^3 - 12t^2 + 12t + 13).$$

The simple eigenvalue 2 therefore has codegree $2p'(2)/q(2) = 10$. The cubic factor is irreducible. Its three eigenvalues are simple, so each corresponding character and codegree is defined over its cubic field. The codegree also lies in K and hence is rational. Galois conjugacy makes the three codegrees equal to one positive integer h. The reciprocal-codegree identity then reads

$$\frac{1}{2} + \frac{2}{180} + \frac{1}{10} + \frac{3}{h} = 1,$$

forcing $h = 54/7$, a contradiction.

The supplied verifier rereads the source census, constructs the full matrix domains, reruns the enumeration, and checks every polynomial gcd and rational projection. It uses no floating-point computation and passes unchanged with Python optimization enabled. □

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